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(54) **VISIBLE LIGHT SOURCE HAVING A
LOW-DENSITY SET OF LIGHT-EMITTING
ELEMENTS AND BACKSIDE OPTICAL
ATTENUATOR**

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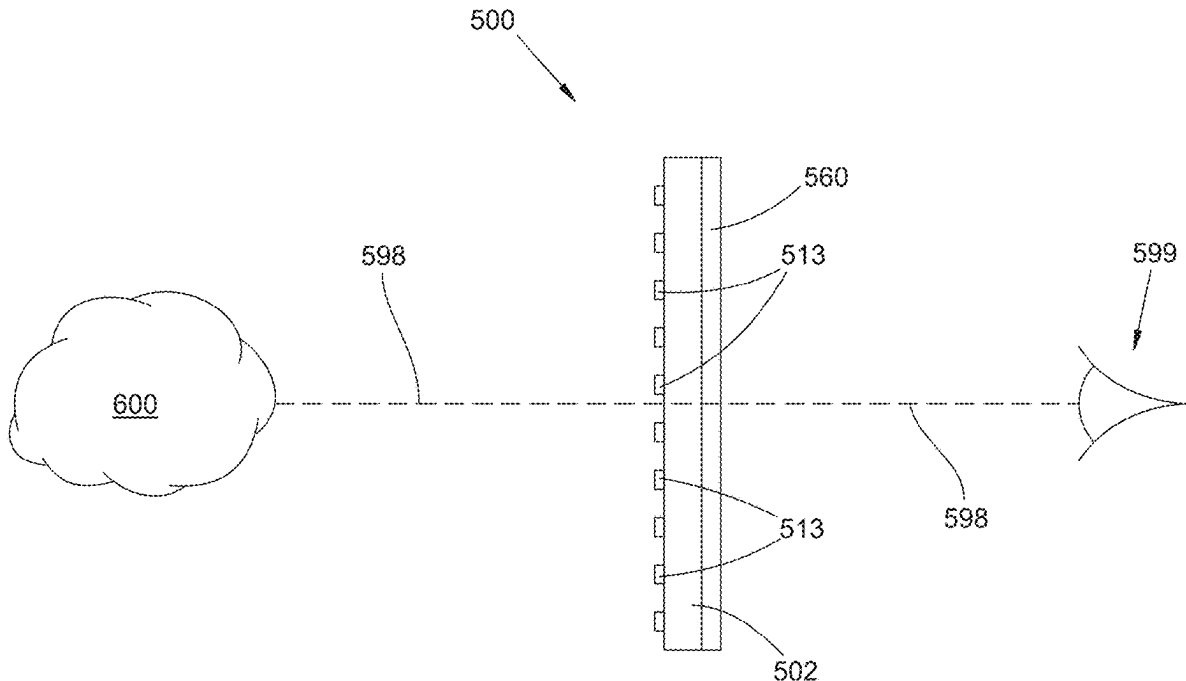
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(57) **ABSTRACT**

An inventive light source includes a transparent substrate, light-emitting elements on or within the substrate, and an optical attenuator structure. The light-emitting elements include inorganic microLEDs that generate and emit visible output light to propagate out-of-plane in a first direction away from a first surface of the substrate. The light-emitting elements are sufficiently small, and occupy a sufficiently small fraction of area, to enable visual observation of a scene through the substrate along a sight line that passes through the set of light-emitting elements. The optical attenuator structure attenuates or blocks transmission of the visible output light that propagates in a second direction opposite the first direction, and transmits enough visible light to enable visual observation of the scene along the sight line through the optical attenuator structure and through the set of light-emitting elements.



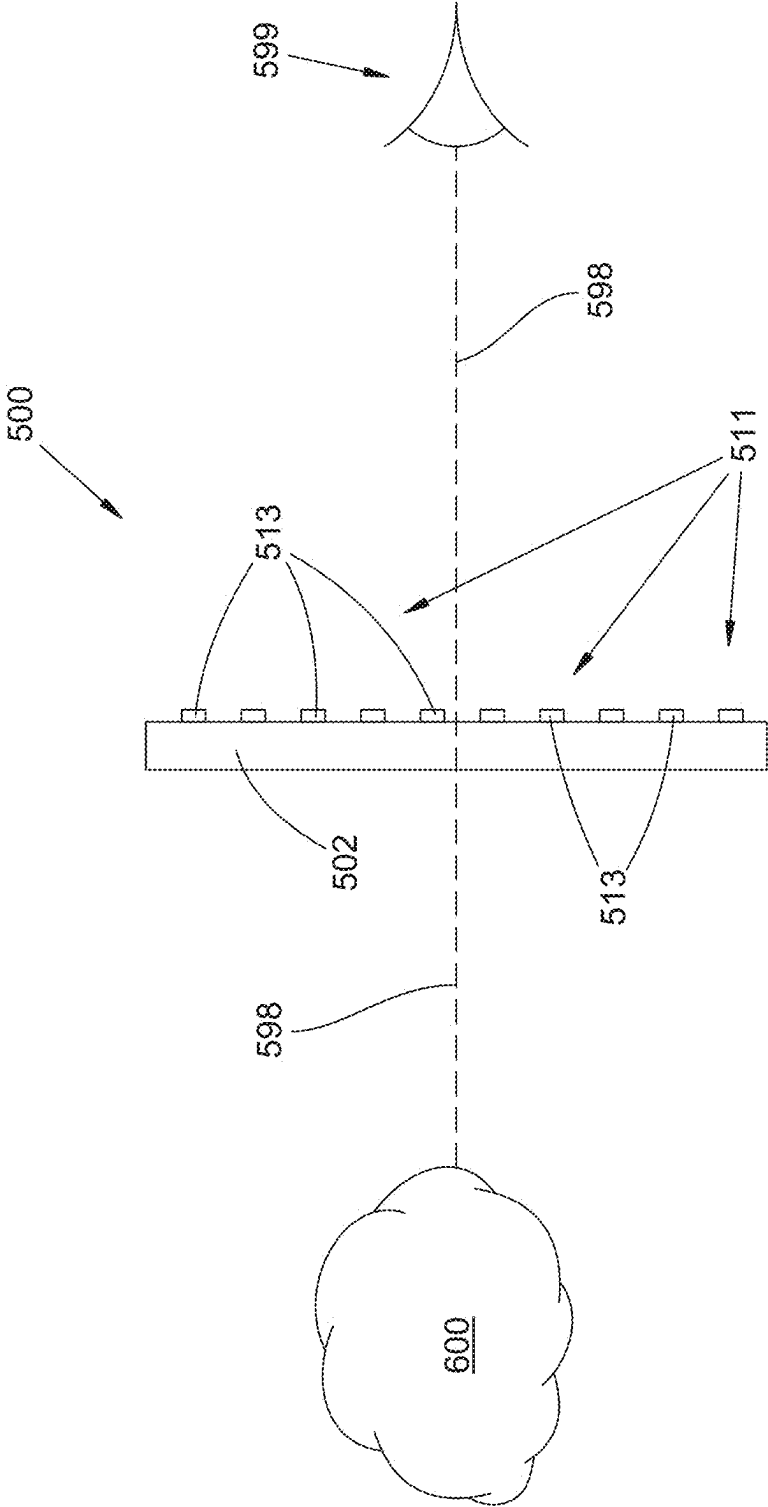
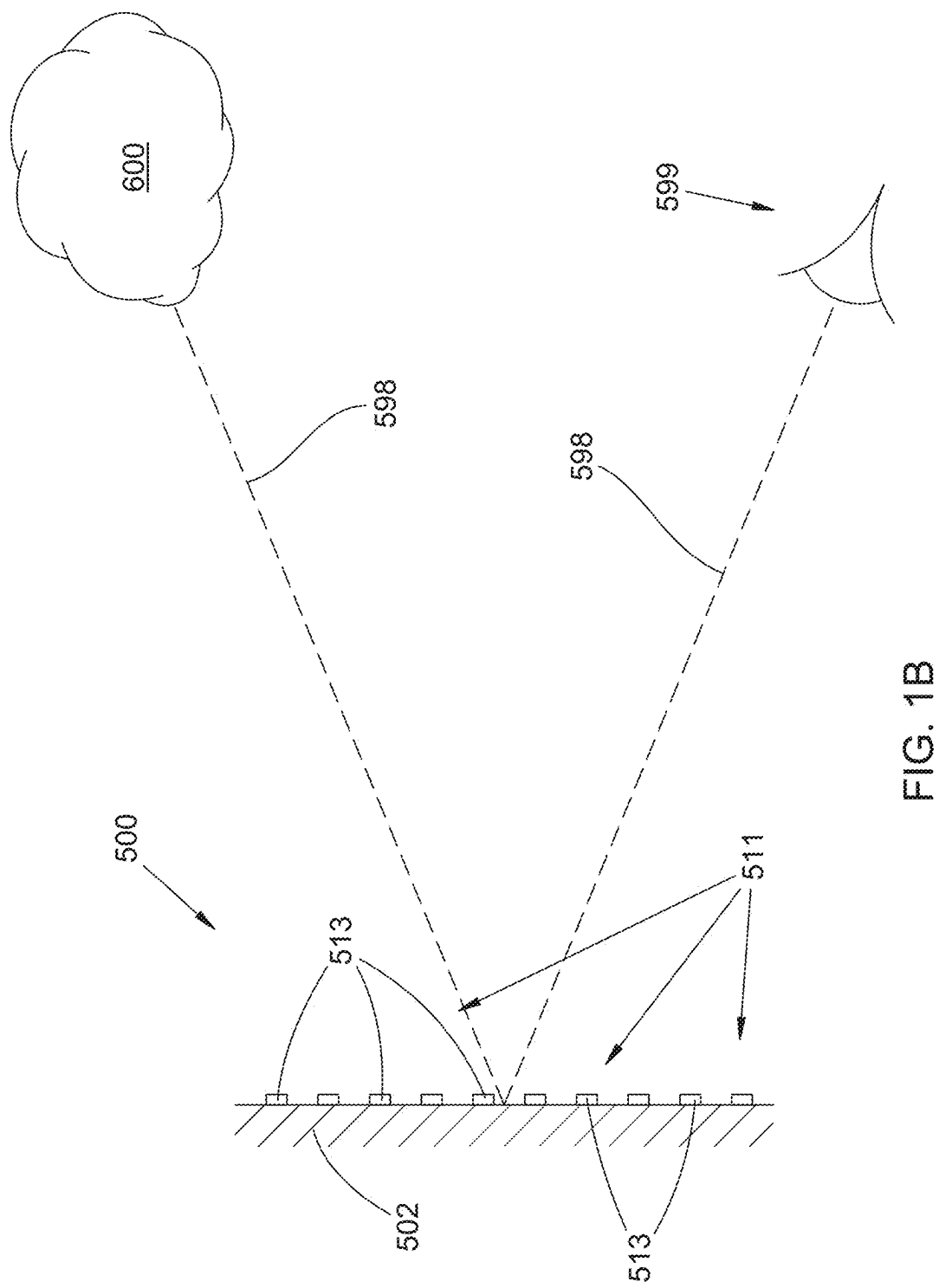


FIG. 1A



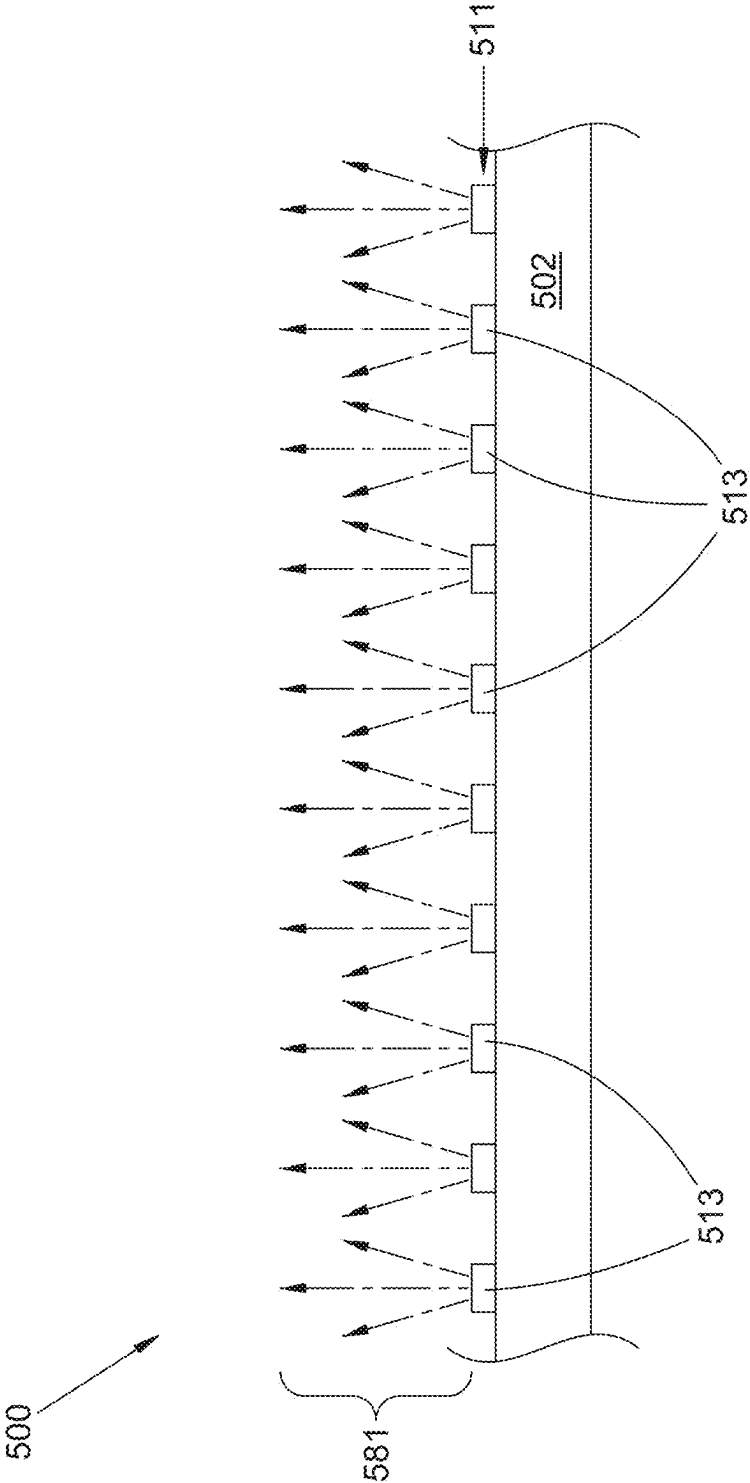


FIG. 2

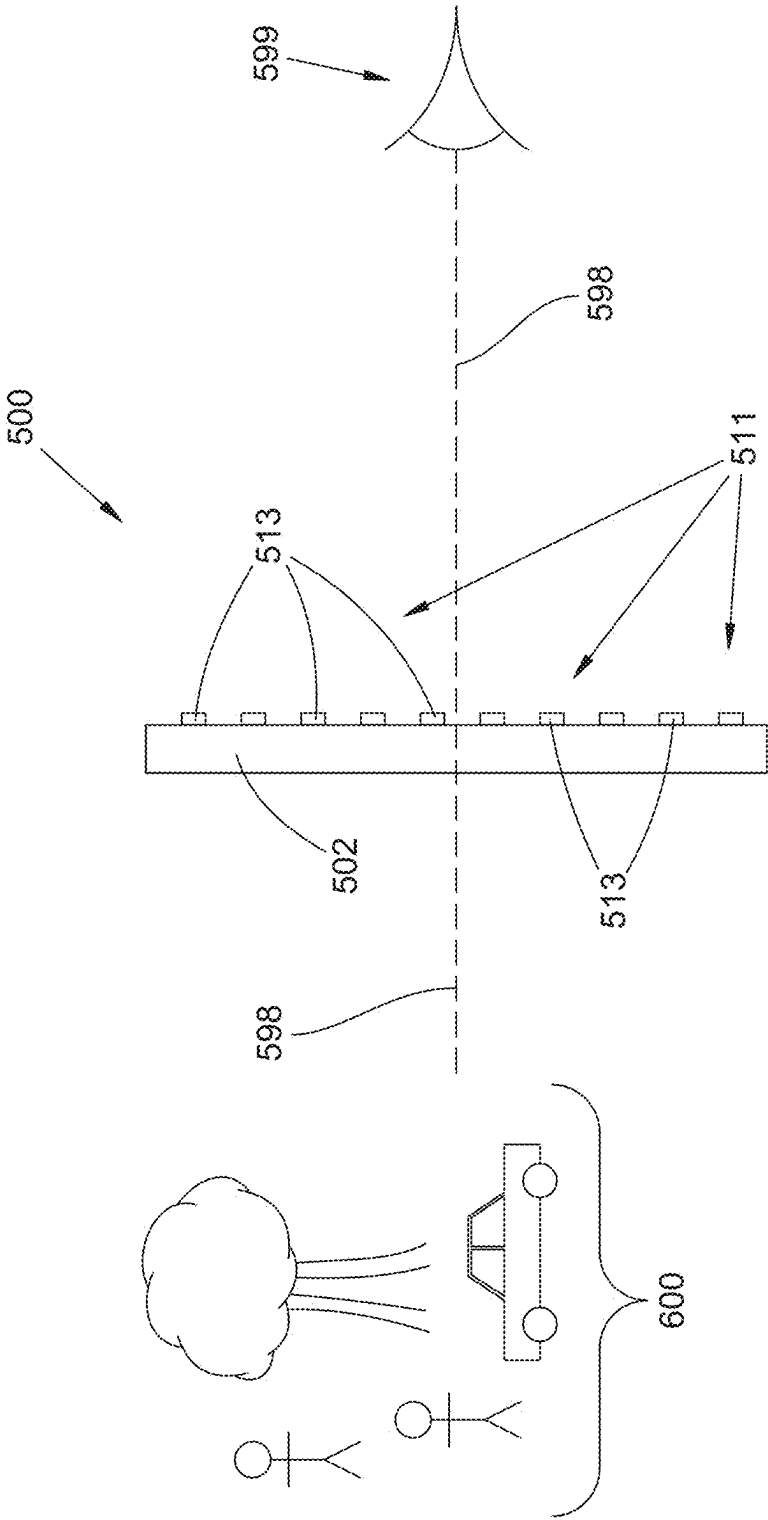


FIG. 3A

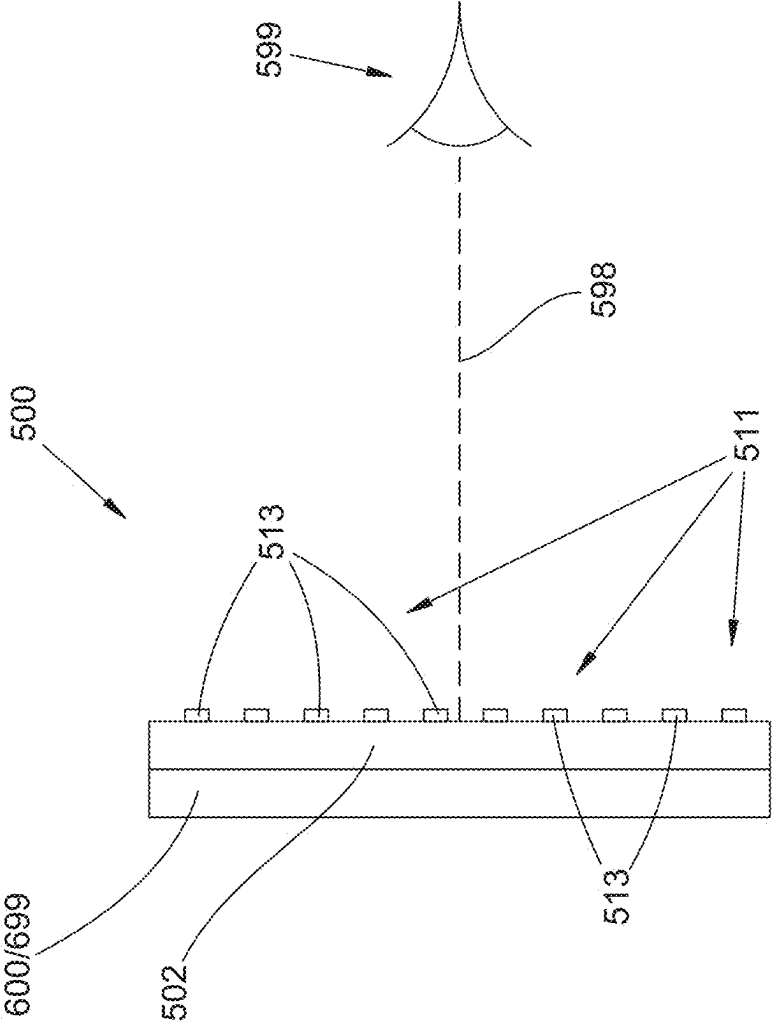


FIG. 3B

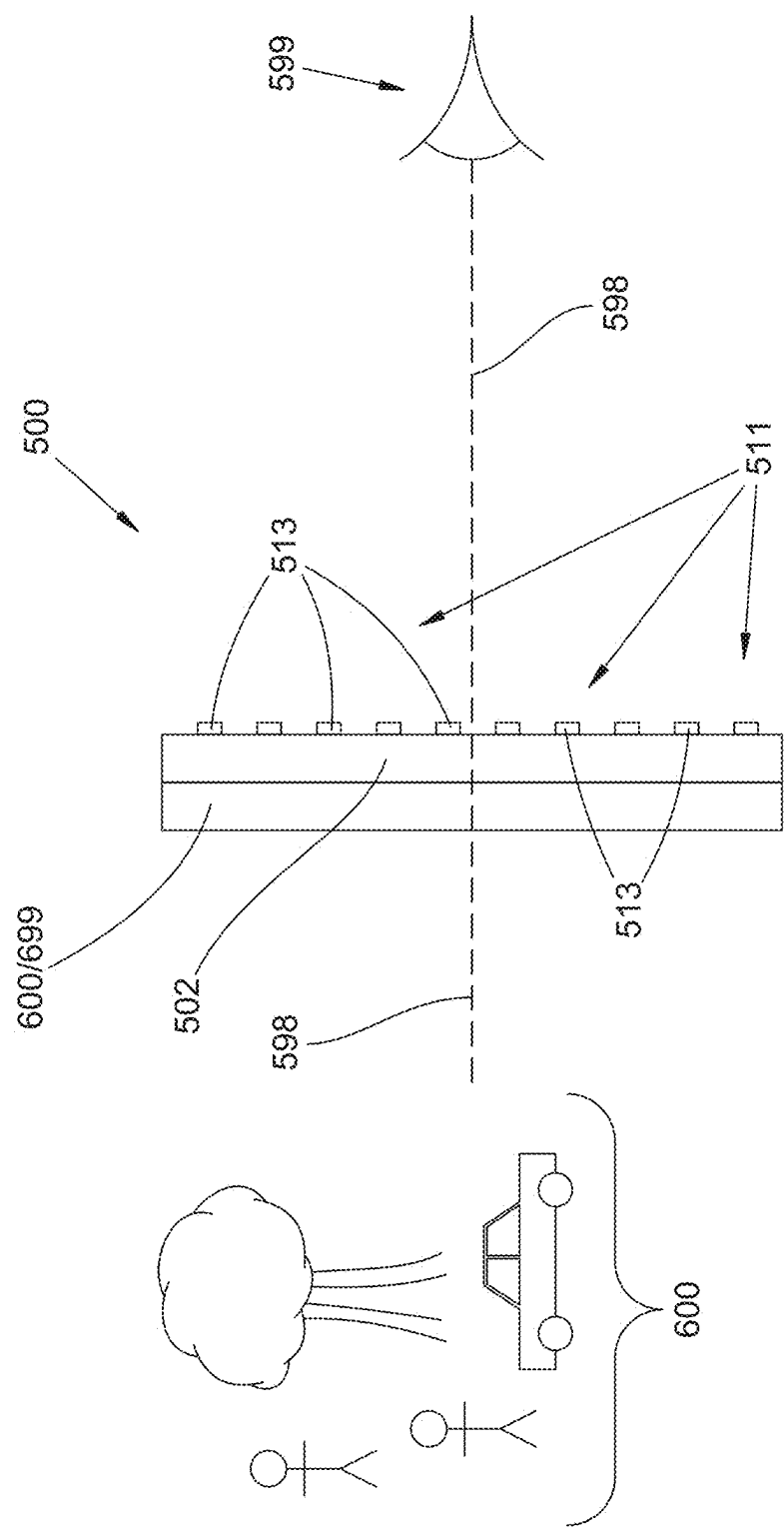


FIG. 3C

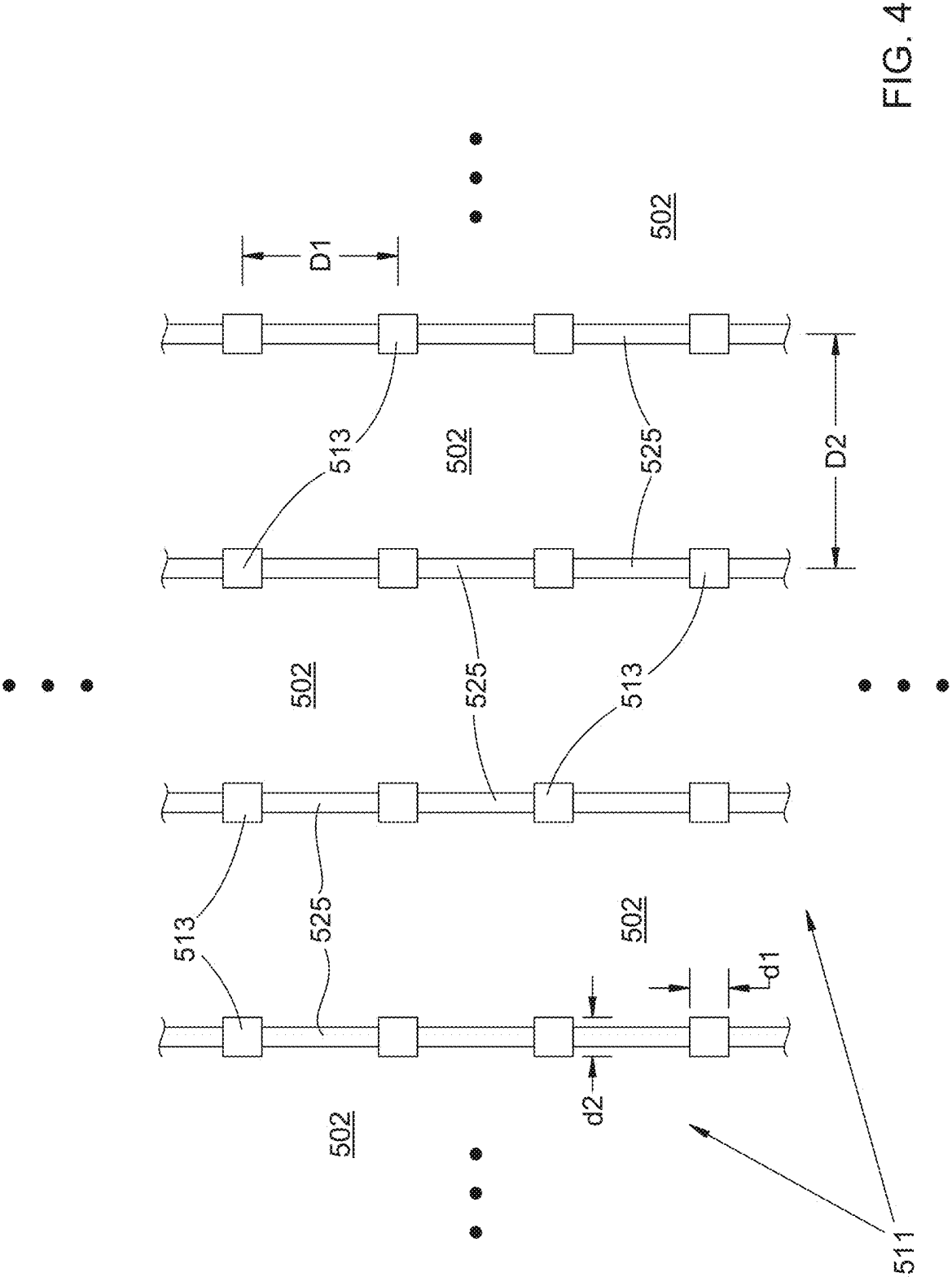


FIG. 4

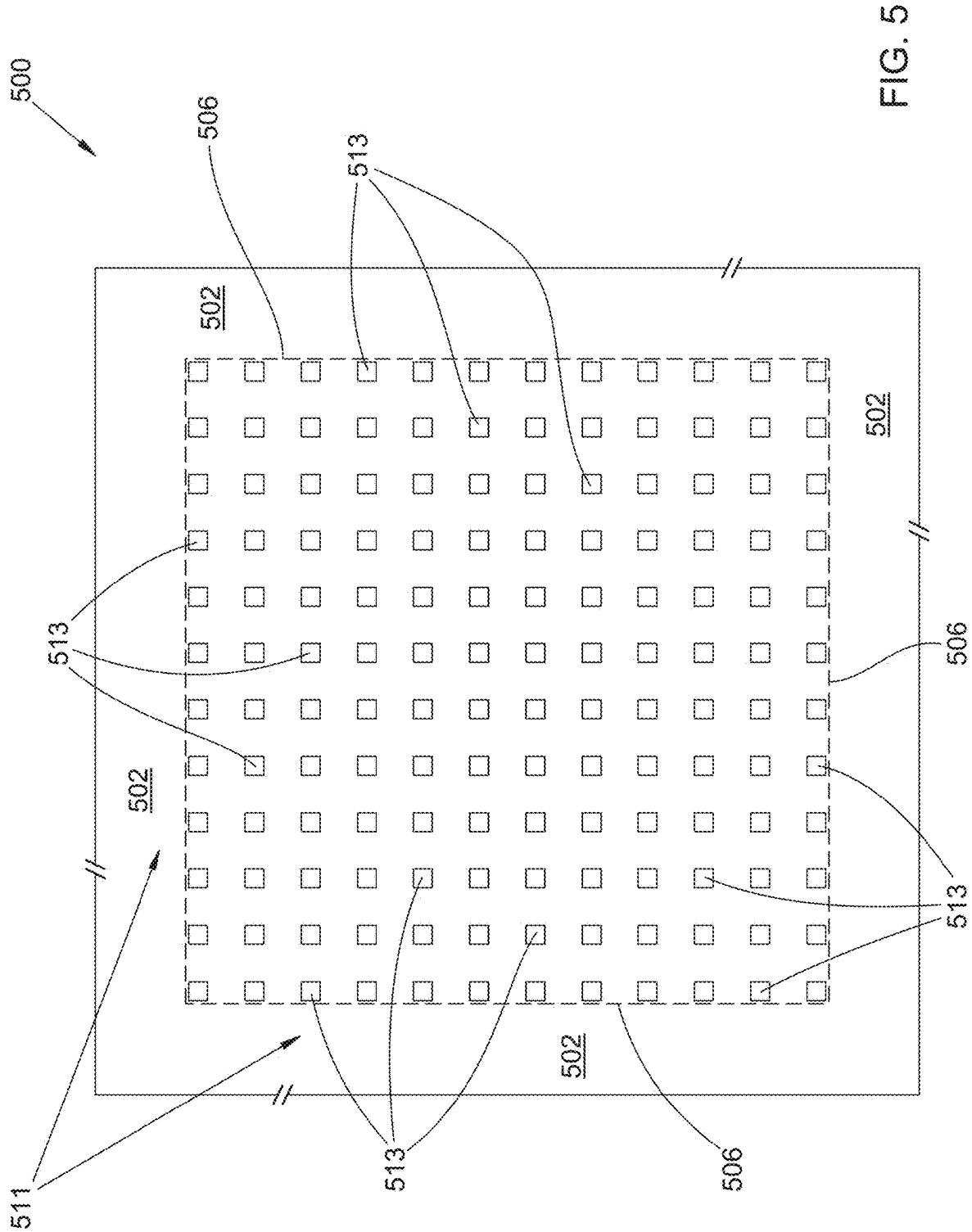


FIG. 5

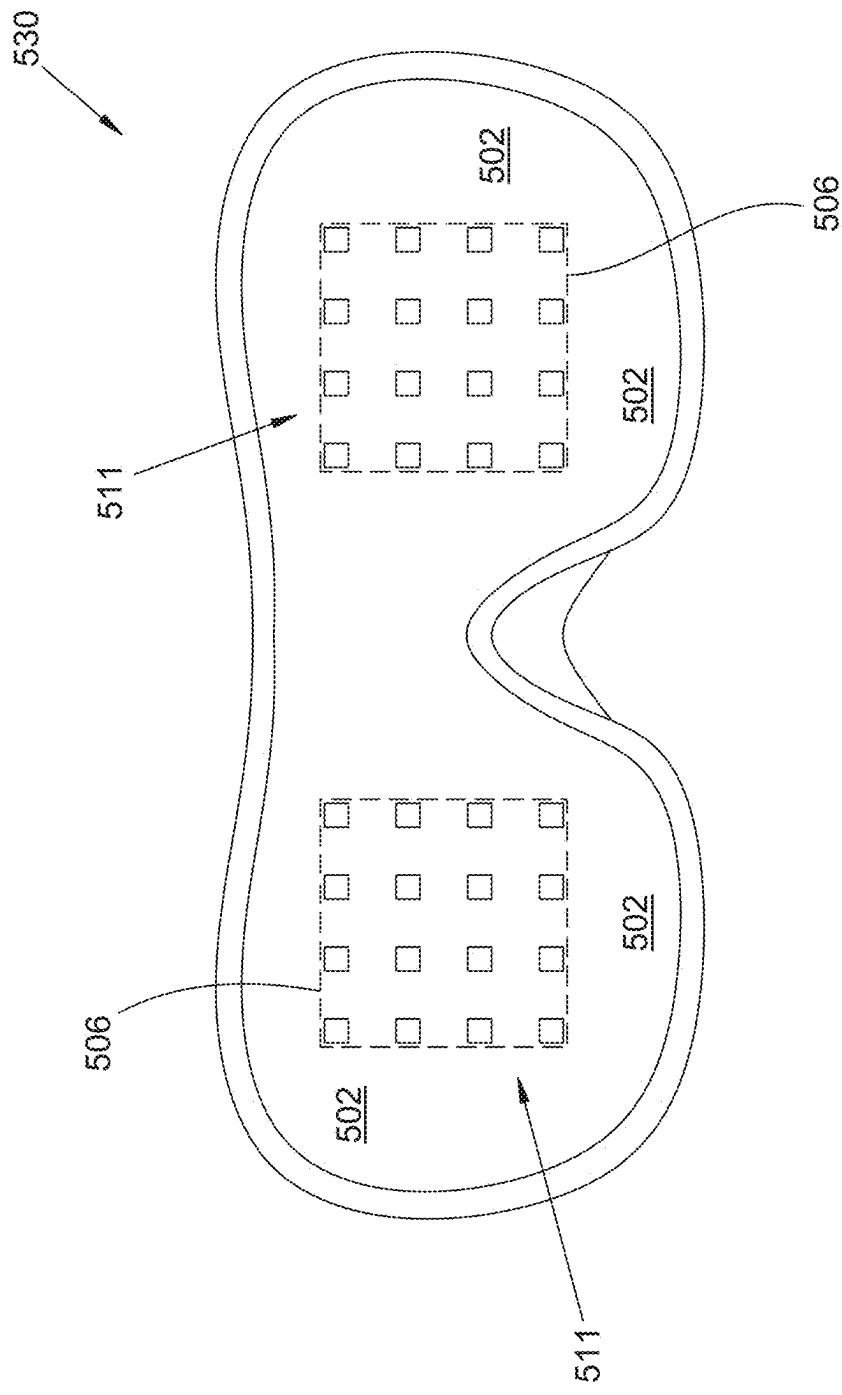


FIG. 6

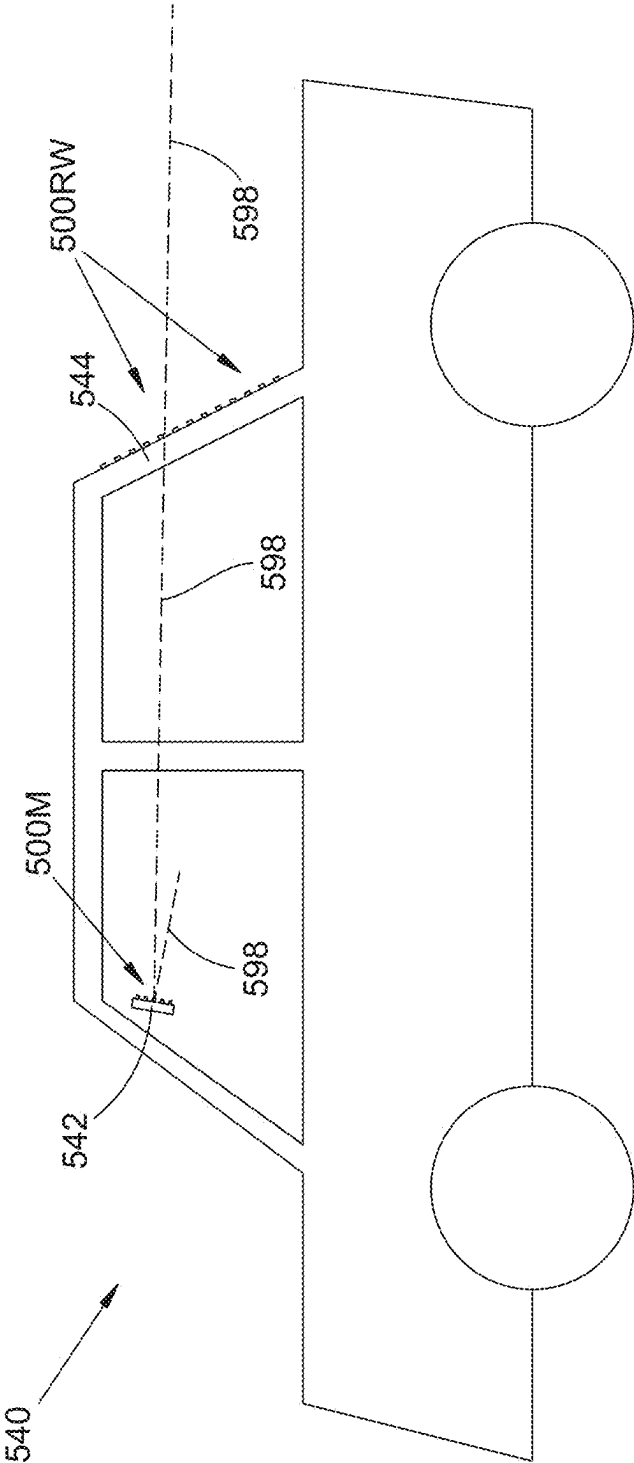


FIG. 7

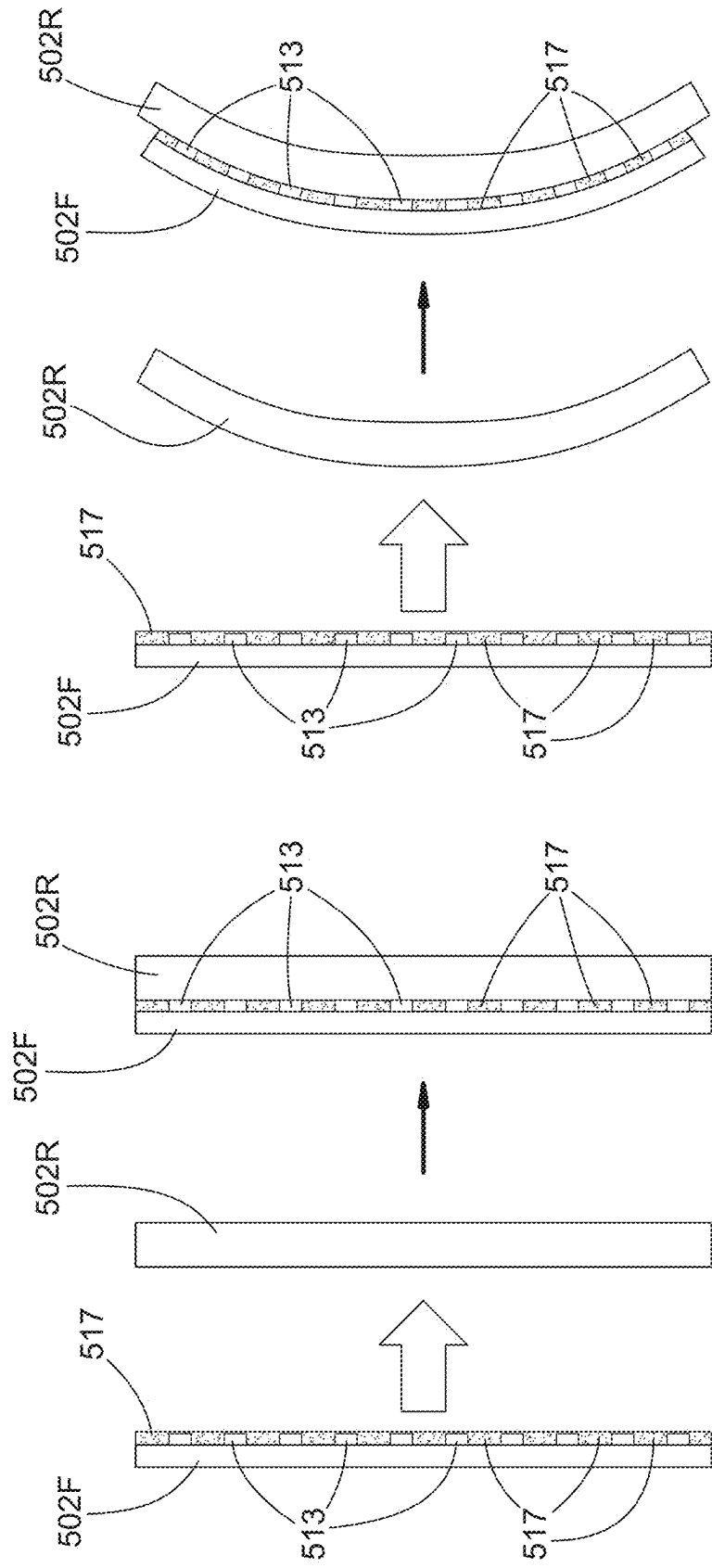


FIG. 8B

FIG. 8A

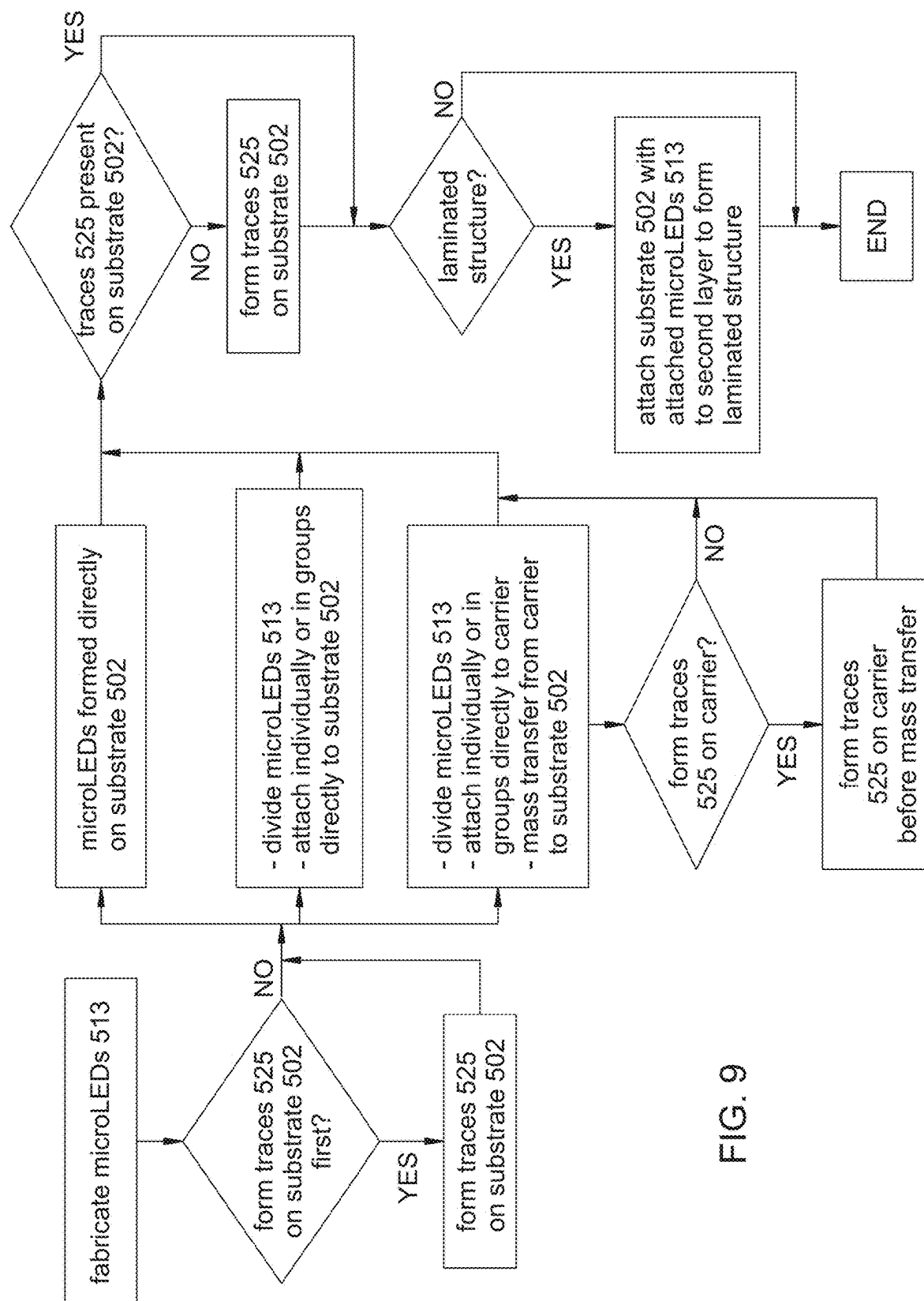
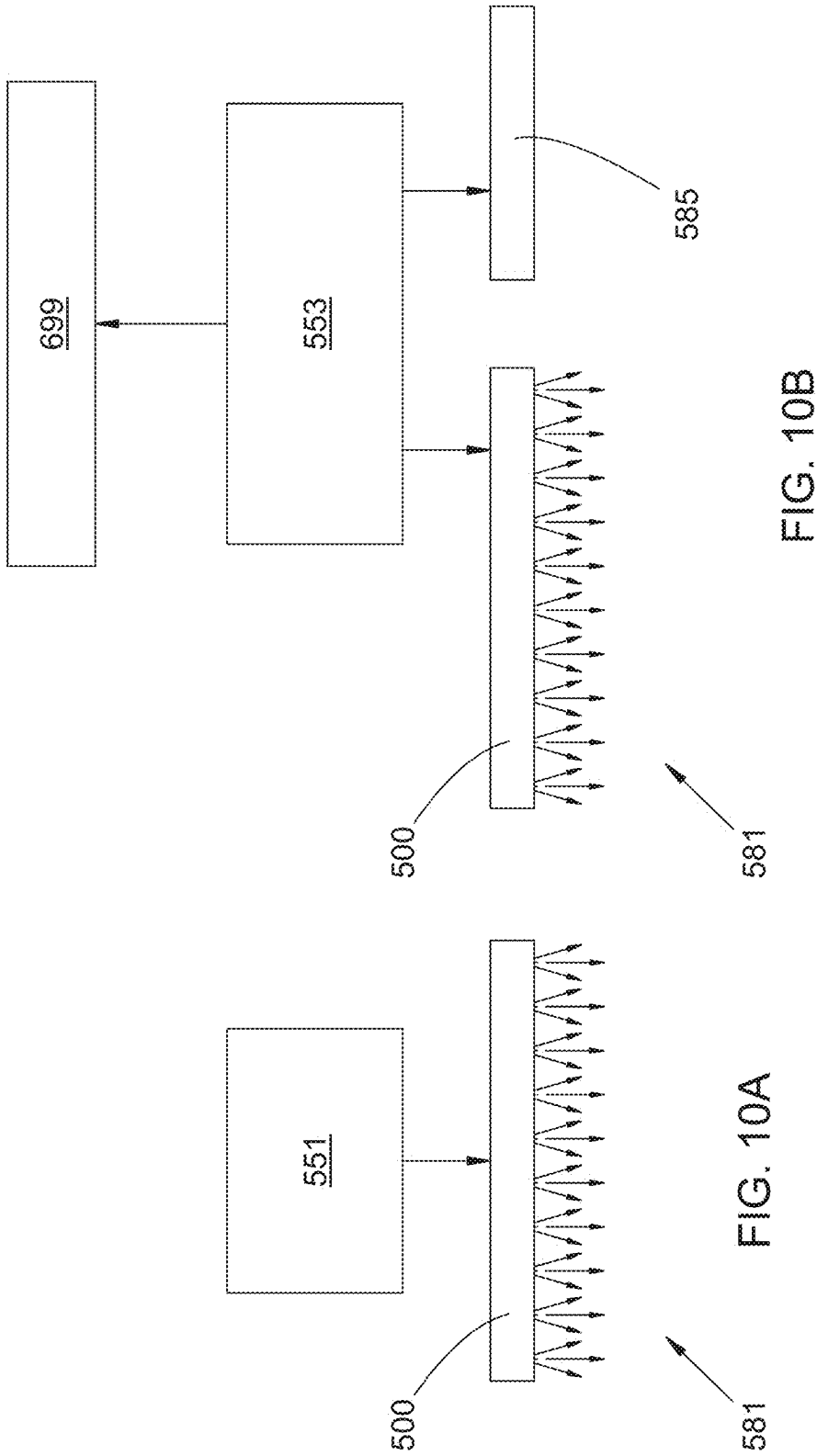


FIG. 9



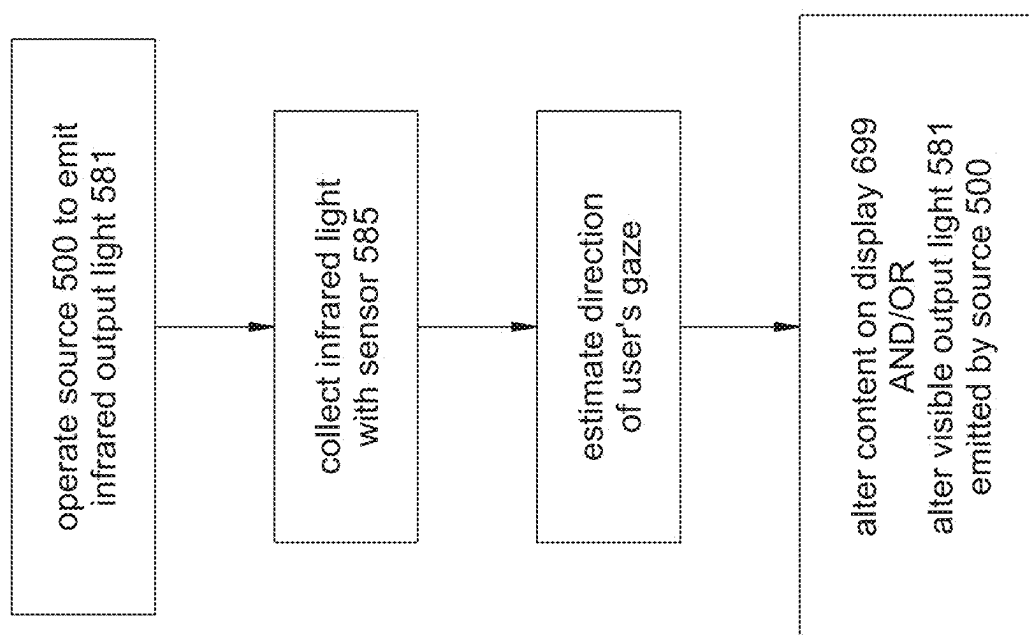


FIG. 11

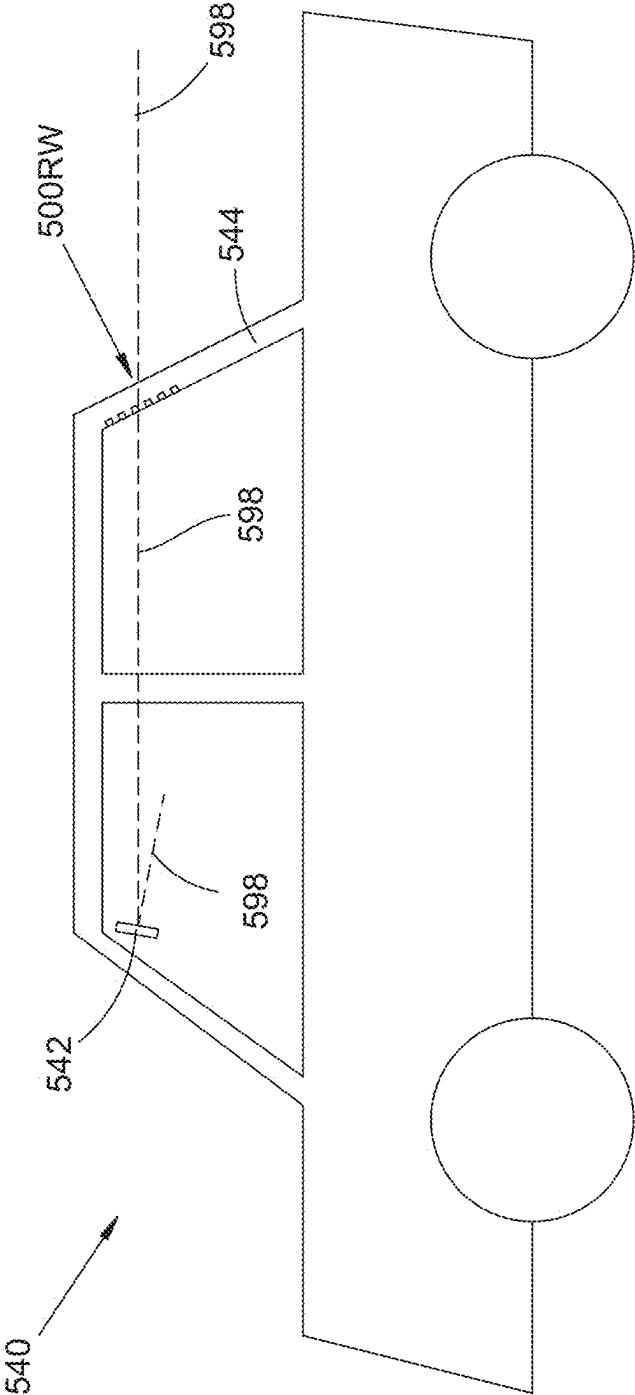


FIG. 12

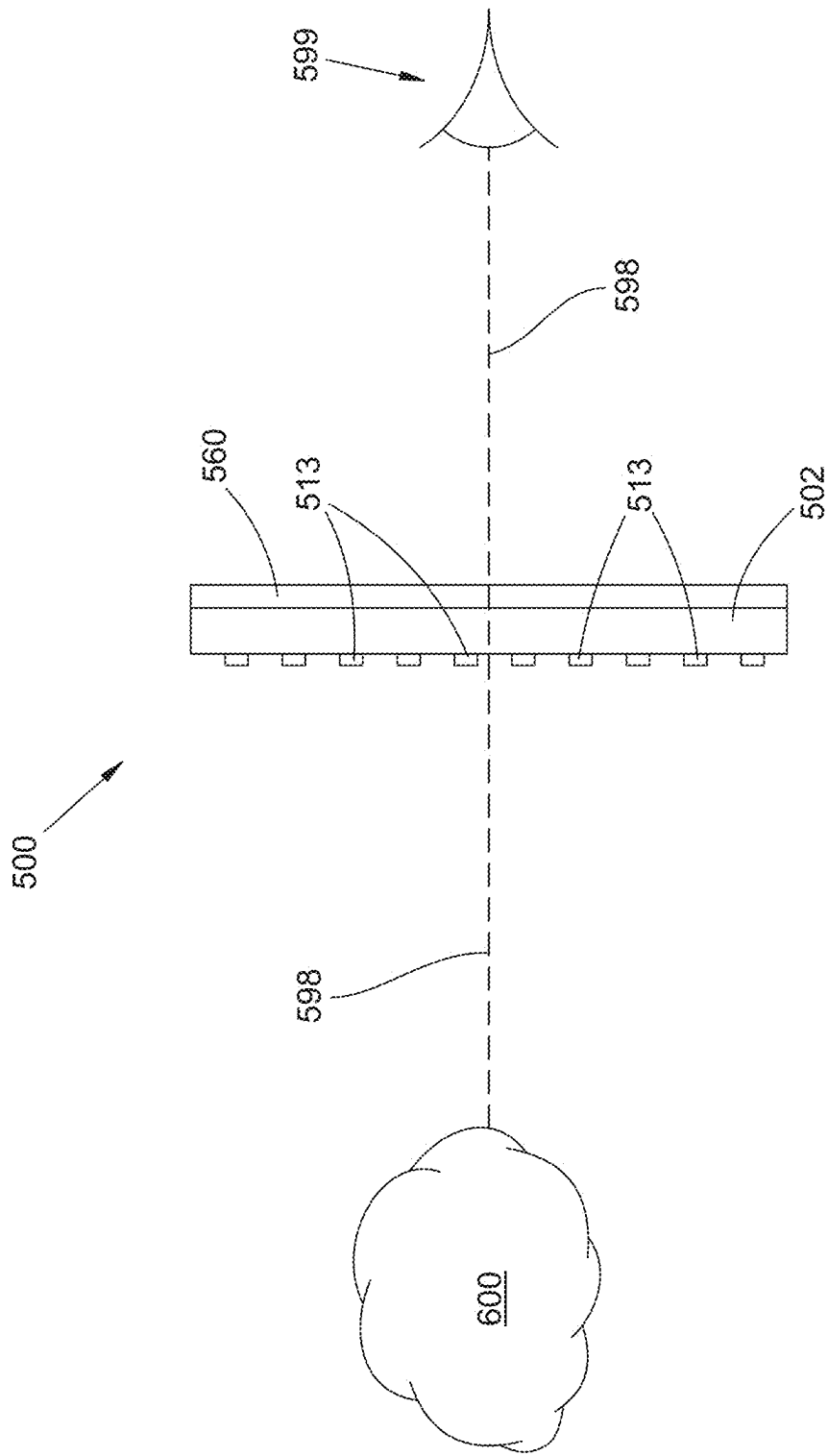


FIG. 13

**VISIBLE LIGHT SOURCE HAVING A
LOW-DENSITY SET OF LIGHT-EMITTING
ELEMENTS AND BACKSIDE OPTICAL
ATTENUATOR**

PRIORITY CLAIM

[0001] This application is a continuation of International App. No. PCT/US2023/083626 entitled “Visible light source having a low-density set of light-emitting elements and backside optical attenuator” filed 12 Dec. 2023 in the name of Hisashi Masui, which in turn claims priority of U.S. provisional App. No. 63/433,420 entitled “Visible light source having a low-density set of light-emitting elements and backside optical attenuator” filed 16 Dec. 2022 in the name of Hisashi Masui; both of said applications are incorporated herein by reference in their entireties.

FIELD OF THE INVENTION

[0002] The field of the present invention relates to light sources. In particular, a “line-of-sight” or “look-through” light source is disclosed that includes a low-density set of light-emitting elements and a backside optical attenuator.

SUMMARY

[0003] An inventive light source includes a substrate, a set of multiple light-emitting elements, and an optical attenuator structure. The substrate is transparent for visible light, and the light-emitting elements are positioned on or within the substrate. Each light-emitting element comprises one or more inorganic microLEDs that are arranged to generate and emit visible output light to propagate away from a first surface of the substrate primarily in a first direction out-of-plane relative to a corresponding localized area of the substrate around that light-emitting element. Each light-emitting element of the set is sufficiently small in at least one transverse dimension, and the light-emitting elements occupy a sufficiently small fraction of an areal extent of the set, so as to enable visual observation of a scene through the substrate along a sight line that passes through the set of light-emitting elements. The optical attenuator structure attenuates or blocks transmission of any of the visible output light that propagates in a second direction opposite the first direction, and transmits enough visible light to enable visual observation of the scene along the sight line through the optical attenuator structure and through the set of light-emitting elements.

[0004] Objects and advantages pertaining to sets of light-emitting elements may become apparent upon referring to the example embodiments illustrated in the drawings and disclosed in the following written description or appended claims.

[0005] This Summary is provided to introduce a selection of concepts in a simplified form that are further described below in the Detailed Description. This Summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to be used as an aid in determining the scope of the claimed subject matter.

BRIEF DESCRIPTION OF THE DRAWINGS

[0006] FIGS. 1A and 1B illustrate schematically transmissive and reflective geometries, respectively, for a low-density set of light-emitting elements.

[0007] FIG. 2 is a schematic cross-sectional view of a low-density set of light-emitting elements on a substrate.

[0008] FIGS. 3A through 3C illustrate schematically observation of actual, virtual, and mixed scenes, respectively, through a low-density set of light-emitting elements.

[0009] FIG. 4 illustrates schematically a plan view of a portion of a low-density set of light-emitting elements and electrical traces on a substrate.

[0010] FIG. 5 illustrates schematically a plan view of a low density set of light-emitting elements on a substrate.

[0011] FIG. 6 illustrates schematically low-density sets of light-emitting elements on eyewear.

[0012] FIG. 7 illustrates schematically low-density sets of light emitting elements on a vehicle.

[0013] FIGS. 8A and 8B illustrate schematically a process for attaching a flexible polymer layer with a low-density set of light-emitting elements to flat and curved rigid layers, respectively.

[0014] FIG. 9 is a flow diagram representing several methods for making a low-density set of light-emitting elements.

[0015] FIG. 10A illustrates schematically a power supply connected to a low-density set of light-emitting elements. FIG. 10B illustrates schematically a low-density set of light emitting elements connected to a control circuit that is also connected to a photodetector, photodetector array, or image sensor and to an electronic display.

[0016] FIG. 11 is a flow diagram representing an example method of using a low-density set of light-emitting elements.

[0017] FIG. 12 illustrates schematically low-density sets of light emitting elements on a vehicle.

[0018] FIG. 13 illustrates schematically a transmissive geometry for a low-density set of light-emitting elements that includes an optical attenuator structure.

[0019] The embodiments depicted are shown only schematically; all features may not be shown in full detail or in proper proportion; for clarity certain features or structures may be exaggerated or diminished relative to others or omitted entirely; the drawings should not be regarded as being to scale unless explicitly indicated as being to scale. For example, the transverse or vertical dimensions of individual light-emitting elements depicted in the drawings may be exaggerated relative to, e.g., their pitch or spacing, or to the thickness of a substrate or other layers. In particular, although the disclosed light sources are intended to be, e.g., only negligibly visible to the naked eye, in some of the drawings the sizes of light-emitting elements might be unrealistically exaggerated (and therefore quite visible) to enable schematic illustration of their arrangement. In some of the drawings only a few light-emitting elements are shown for illustrative purposes, while an actual light source might have a larger number of light-emitting element. Some schematic illustrations of example structures of various devices or assemblies described herein may be shown with precise right angles and straight lines, but it is to be understood that such schematic illustrations may not reflect real-life process limitations or defects. Such process limitations or defects can cause the features to look not so “ideal” when any of the structures described herein are examined using, e.g., optical microscopy images, scanning electron microscopy (SEM) images, or transmission electron microscope (TEM) images. In such images of real structures, possible processing limitations or defects might be visible,

e.g., not-perfectly straight edges of materials, slanted side walls, tapered vias or other openings, inadvertent rounding of corners, or variations in thicknesses of different material layers. There may be other limitations or defects not listed here that can occur within the field of device fabrication. The embodiments shown are only examples and should not be construed as limiting the scope of the present disclosure or appended claims.

DETAILED DESCRIPTION

[0020] The following detailed description should be read with reference to the drawings, in which identical reference numbers refer to like elements throughout the different figures. The drawings, which are not necessarily to scale, depict selective examples and are not intended to limit the scope of the disclosed inventive subject. The detailed description illustrates by way of example, not by way of limitation, the principles of the disclosed inventive subject matter.

[0021] Subject matter disclosed herein may be related to subject matter disclosed in one or more of: (i) U.S. non-provisional application Ser. No. 17/985,886 filed 13 Nov. 2022, (ii) U.S. non-provisional application Ser. No. 17/985,888 filed 13 Nov. 2022, (iii) U.S. non-provisional application Ser. No. 17/985,892 filed 13 Nov. 2022, (iv) U.S. non-provisional application Ser. No. 17/985,893 filed 13 Nov. 2022, (v) U.S. non-provisional application Ser. No. 17/985,894 filed 13 Nov. 2022, and (vi) U.S. non-provisional application Ser. No. 17/985,897 filed 13 Nov. 2022. Said applications are all incorporated herein by reference in their entireties.

[0022] There are a number of scenarios in which it is necessary or desirable to track position or movement of eyes or head of a human observer. For example, such tracking can be employed in a vehicle to monitor the attention of a driver. In other examples, such tracking can be employed to estimate the direction of a user's gaze in a visualization system (e.g., a virtual reality (VR) system or an augmented reality (AR) system), or in a static or dynamic electronic display (e.g., a video or data monitor, electronic signage, electronic out-of-home advertising, and so forth). Such eye-tracking information can in turn be used to guide generation or alteration of audiovisual content presented to the user in the visualization system or on the electronic display. Typically, one or more light sources are used to illuminate the user's eyes or face (often using infrared light to avoid distracting the user), and a sensor or detector (e.g., an imaging sensor) is employed to detect light reflected or scattered from the user's eye or face. Processing of the resulting signals yields an estimate of the direction of the user's gaze.

[0023] Previous eye-tracking systems have included illuminating light sources located at oblique angles relative to the user's line of sight. One common arrangement includes a set of light sources positioned on the peripheral frame of a set of AR/VR eyewear (e.g., glasses or goggles) or other wearable optical assembly. In some systems such AR/VR eyewear might be provided solely for providing eye tracking, with visual content of the AR/VR environment delivered using a separate screen or display (e.g., a head-mounted display, faceplate, or visor, or a display screen or window). In other AR/VR systems the eyewear with the eye-tracking light sources can also be used for delivering visual content of the AR/VR system, by forming images on the lenses or windows of the eyewear.

[0024] The oblique illumination provided by previous eye-tracking hardware is disadvantageous for several reasons. The locations of the sources on the peripheral frame of the eyewear place them at greater distance from the eye, so that correspondingly higher optical power might be needed to produce an adequately detectable signal. The oblique incidence angle of the light on the eye might also reduce the amount of light returned to the sensor or detector, also requiring higher-output light sources or more sensitive sensors or detectors. Accuracy of the estimated direction of the user's gaze might be limited by use of only off-axis light to illuminate the eye. Therefore it would be desirable to provide an illuminating light source that can be placed in an observer's line of sight so as to more directly illuminate the observer's eyes or face, while also not substantially interfering with observation of a scene along the observer's line of sight through the light source. Similarly, in an automotive setting it would be desirable to place a light source for illuminating the driver's eyes or face within the driver's line of sight without interfering substantially with the driver's vision. Such a "line-of-sight" or "look-through" light source could also be advantageously employed in other scenarios or for other purposes, some of which are described below.

[0025] Accordingly, an inventive light source **500** comprises a substrate **502** and a set **511** of multiple discrete light-emitting elements **513**. The substrate **502** is transparent or reflective for visible light (e.g., light having a vacuum wavelength between about 400 nm and about 700 nm). "Transparent" denotes sufficient transmission of visible light for the light source **500** to operate as needed in a given context. "Reflective" similarly denotes sufficient reflection of visible light for the light source **500** to operate as needed in a given context. Reflectivity of the substrate **502** can be provided in some instances by the inherent reflectivity of a surface of the substrate material, or in other instances can be imparted by a suitable reflective coating (e.g., metallic, multilayer dielectric, and so forth) on a surface of the substrate **502**. The general transmissive and reflective geometries are illustrated schematically in FIGS. 1A and 1B, respectively. Much of the ensuing description applies generally to both geometries, except where only one or the other geometry is specifically mentioned. In some examples the substrate **502** can be partially transmissive and partially reflective.

[0026] The set **511** of multiple discrete light-emitting elements **513** is positioned on or within the substrate **502**, with each light-emitting element **513** being arranged to emit output light **581** out-of-plane relative to a corresponding localized area of the substrate **502** surrounding that light-emitting element **513** (e.g., illustrated schematically in FIG. 2). In other words, the light source **500** is not intended to function by lateral propagation of light within the substrate **502**, but by generation and emission of output light **581** by the light-emitting elements **513** to propagate away from one or both surfaces of the substrate **502**. Note that in a system incorporating the light source **500**, it is possible that other components of that system could operate by lateral propagation of light within the substrate **502** or within another substrate on which the substrate **502** is positioned. In many examples output light **581** emitted by each light-emitting element **513** propagates in an angular distribution centered about the surface normal vector of the substrate **502** at the location of that light-emitting element **513**. The set **511** can be referred to as low-density or sparse, meaning that each

light-emitting element **513** is sufficiently small in one or both transverse dimensions (i.e., dimensions locally parallel to the substrate surface), and that the light-emitting elements **513** of the set **511** occupy a sufficiently small fraction of an areal extent **506** of the set **511**, so as to enable visual observation of a scene **600** through or reflected by the substrate **502** along a sight line **598** that passes through the set of light-emitting elements **511**.

[0027] In some examples (e.g., as in the example of FIG. 2) at least some of light-emitting elements **511** can be positioned on one surface of the substrate **502** and at least some of the output light **581** from those elements **511** can propagate away from that same surface of the substrate **502**. In some examples (not shown) at least some of the light-emitting elements **511** can be positioned on one surface of the substrate **502** and at least some of the output light **581** from those elements **511** can propagate through the substrate **502** to propagate away from the opposite surface thereof. In some examples (not shown) at least some of the light-emitting elements **511** can be positioned within the substrate **502**. In some such examples all of the output light **581** of those elements **511** can propagate through a portion of the substrate **502** and away from either one surface of the substrate **502** or the other surface but not both; in other such examples one portion of the output light **581** of those elements **511** can propagate through a portion of the substrate **502** and away from one surface thereof, while another portion of the output light **581** can propagate through a portion of the substrate **502** and away from the opposite surface thereof. In examples wherein the output light is intended to propagate in only one direction relative to the substrate, there can be unwanted propagation of some output light in the other direction. Mitigation of such unwanted propagation is discussed further below.

[0028] In some examples the scene **600** can be a region of actual space that is viewed by a human observer **599** along the human observer's line of sight **598** through the substrate **502** (e.g., as in FIG. 3A) or reflected from the substrate **502**; the region of space constituting the scene **600** can include any one or more structures, objects, or people located in that region of space (as illustrated schematically in FIG. 3A). In some examples the scene **600** can be an image (static or moving) that is formed on an electronic display **699** (e.g., computer, tablet, television, or phone screen; electronic signage or billboard display; VR eyewear or headset) within the line of sight **598** of the human observer **599** and that is viewed by the human observer **599** through the substrate **502** (e.g., as illustrated schematically in FIG. 3B) or reflected from the substrate **502**. In some of those examples the display **699** can be offset separated from the light source **500**; in others of those examples the light source **500** can be positioned on the surface of the display **699** (e.g., as in FIG. 3B) or be incorporated into the display **699**. In a specific example the display **699** can include suitably modulated edge light sources that launch light to propagate laterally within the substrate **502** to create images seen by the human observer. In some examples (e.g., AR eyewear or headset; transparent window) the scene **600** can include both a region of actual space and a superimposed image formed on a display **699** through which the region of actual space is viewed (as illustrated schematically in FIG. 3C). Each of those scenarios is discussed in further detail below.

[0029] In some examples the light-emitting elements **513** of the set **511** can be small enough and spread far enough

apart on the substrate **502** so that, with the light-emitting elements **513** in an off state or emitting only infrared light, the set **511** does not substantially interfere with visual observation of the scene **600** by a naked eye of a human observer **599**, through the substrate **502** or reflected by the substrate **502**, along the line of sight **598** that passes through the set **511**. Note that the "naked eye" of the human observer might include ordinary vision-corrective lenses, such as eyeglasses or contact lenses, that are used to improve daily vision in people having refractive errors. In some examples the light-emitting elements **513** of the set **511** can be small enough and spread far enough apart on the substrate **502** so that, when they are in an off state or emitting only infrared light, they are effectively invisible to the naked eye of the human observer **599**.

[0030] In some examples, the light-emitting elements **513**, although small and spread apart, when in an off state or emitting only infrared light they may nevertheless be visible to the naked eye. In such examples, if a transparent substrate **502** with a low-density set **511** of light-emitting elements **513** were held up to a solid white background, or if a light were shone on the set **511** at an oblique angle, an attentive human observer might discern the presence of the set **511** (e.g., by slight discoloration or grainy appearance of the substrate **502**, or light scattered from the light-emitting elements **513**), even with the light-emitting elements **513** in an off state or emitting only infrared light. But like dust on a television or computer screen, which might be somewhat apparent when the screen is dark, and more readily apparent when a light shines obliquely across the dark screen, that dust can easily go unnoticed by a human observer when images are displayed on the screen, effectively disappearing from view. Similarly, dust on eyeglasses might be visible with the eyeglasses held up to a light to look for the dust, but can easily go unnoticed by the wearer observing a scene through those same dusty eyeglasses. That same phenomenon is exploited by the low-density arrangement of the set **511** of light-emitting elements **513** of the light source **500**. Such a set **511** in some instances can be characterized as being only negligibly visible, or as resembling dust on the substrate **502**.

[0031] The light source **500** can include multiple electrically conductive traces **525** on the substrate **502** that are arranged for providing electrical drive current to the light-emitting elements **513** of the set **511** (illustrated schematically in FIG. 4). In some examples the conductive traces **525** can include transparent traces and can include one or more materials among indium tin oxide (ITO), indium zinc oxide (IZO), or one or more other transparent conductive oxides (TCOs). Such transparent traces can be sufficiently transparent, or sufficiently narrow and spaced sufficiently far apart, so that they also (like the light-emitting elements **513**) enable visual observation of the scene **600** through or reflected by the substrate **502** along the sight line **598** that passes through the set of electrically conductive traces **525**. In some examples the conductive traces **525** can include metallic traces and can include one or more materials among aluminum, silver, gold, or one or more other metals or metallic alloys. Such metallic traces can be sufficiently narrow and spaced sufficiently far apart so that they also (like the light-emitting elements **513**) enable visual observation of the scene **600** through or reflected by the substrate **502** along the sight line **598** that passes through the set of electrically conductive traces **525**. The conductive traces

525 of either or both types in some instances can be characterized as only negligibly visible, as resembling dust on the substrate **502**, or as not substantially interfering with visual observation of the scene **600** by the naked eye **599** of the human observer, through or reflected by the substrate **502**, along the line of sight **598** that passes through the traces **525**.

[0032] In some examples all of the light-emitting elements **513** of the set **511** can be connected to a source of electrical current (e.g., a power supply **551** as in FIG. **10A**, or a control circuit **553** as in FIG. **10B**) so as to operate in unison. In some such examples one or more or all of the traces **525** can constitute a single conductive element so that multiple light-emitting elements **513** can be connected, e.g., in series. In other such examples one or more or all of the traces **525** can constitute two or more independent conductive elements so that multiple light-emitting elements **513** can be connected, e.g., in parallel. In some examples one or more or all of the light-emitting elements **513** can be connected to a control circuit **553** so as to be operable independently of one or more or all of the other light-emitting elements **513** of the set **511**. In some such examples one or more or all of the traces **525** can constitute two or more independent conductive elements to enable such independent operation.

[0033] In general the light-emitting elements **513** can be of any suitable type, composition, or arrangement. In some examples the light-emitting elements **513** of the set **511** can include one or more inorganic light-emitting diodes (LEDs). In some examples each such inorganic LED **513** of the set **511** can comprise an inorganic semiconductor LED and can include one or more materials among III-V, II-VI, or Group IV semiconductor materials. In some examples one or more light-emitting elements **513** of the set **511** can include so-called direct-emitting LEDs, in which light generated by radiative recombination of charge carriers at an active layer or region of the LED constitutes the entire light output of the direct-emitting LED. In some examples one or more of the light-emitting elements **513** can include so-called phosphor-converted LEDs that include a wavelength-converting structure, e.g., one or more phosphors that absorb light at the wavelength generated by recombination in the LED and in turn emit light at one or more different, longer wavelengths. Light output of such phosphor-converted LEDs can include only light at the longer wavelength(s), or can include both direct-emitted and phosphor-converted light.

[0034] In some examples one or more of the light-emitting elements **513** (direct-emitting, phosphor-converted, or both types) of the set **511** can emit non-visible light such as infrared light (e.g., light having a wavelength longer than about 750 nm, longer than about 800 nm, or longer than about 840 nm); in some of those examples all of the elements **513** emit only non-visible light such as infrared light. In some examples one or more of the light-emitting elements **513** (direct-emitting, phosphor-converted, or both types) of the set **511** can emit visible light (e.g., light having a wavelength between about 400 nm and about 700 nm or 750 nm); in some of those examples all of the elements **513** emit only visible light. In some examples each light-emitting element **513** of the set **511** can emit light at one or more corresponding wavelengths that differ from one or more corresponding wavelengths of light emitting by at least one other light-emitting element **513** of the set **511** (i.e., the set **511** can be a multicolor set); in some other examples each light-emitting element **513** of the set **511** can emit light the

same one or more wavelengths as light emitted by all other light-emitting elements **513** of the set **511** (i.e., the set **511** can be a monochrome set). In some examples one or more of the light-emitting elements **513** is a unitary element; in some of those examples all of the elements **513** are unitary elements. In some other examples one or more of the light-emitting elements **513** can be arranged as a compound element including two or more independently operable light-emitting subelements (e.g., independently operable LEDs; direct-emitting, phosphor-converted, or both types). The subelements of each compound light-emitting element **513** can emit light at corresponding wavelengths different from one another, so that the overall color of the light emitted by each compound light-emitting element **513** can be varied by varying the relative intensities emitted by each subelement. In some examples all of the light-emitting elements **513** can be arranged as compound elements.

[0035] In some examples each light-emitting element **513** can have a largest transverse dimension (referring to FIG. **4**, $\max(d1, d2)$) that is less than 200 μm , less than 100 μm , less than 50 μm , less than 30 μm , less than 20 μm , less than 10 μm , less than 8 μm , less than 5 μm , or even smaller. LEDs of those sizes can be referred to as microLEDs. The nonzero transverse dimensions **d1** and **d2** of each light-emitting element **513** can be made as small as desired or practicable while still enabling the element **513** to emit light (e.g., for an LED to still function as an LED). The light-emitting elements **513** can be of any suitable shape (e.g., square, rectangular, circular, etc) and need not be the same size in both transverse dimensions; in FIG. **4** the light-emitting elements **513** are shown as being square (if $d1=d2$) or rectangular (if $d1 \neq d2$). In some examples spacing or pitch (center-to-center; referring to FIG. **4**, $\min(D1, D2)$) of the light-emitting elements **513** of the set **511** can be greater than 0.1 mm, greater than 0.2 mm, greater than 0.3 mm, greater than 0.5 mm, greater than 1.0 mm, greater than 2 mm, greater than 3 mm, greater than 5 mm, or even larger. The set can be of any suitable arrangement (e.g., square, rectangular, trigonal, hexagonal, or other regular or periodic array, or an aperiodic, irregular, or random arrangement). If periodic the spacing or pitch need not be the same across the entire array and need not be the same in both transverse dimensions along the surface of the substrate **502**. In FIG. **4** the set **511** is shown as being arranged as a square array (if $D1=D2$) or as a rectangular array (if $D1 \neq D2$). In some examples the set **511** can occupy an area **506** of the substrate **502** (e.g., illustrated schematically in FIGS. **5** and **6**) having a smallest transverse dimension that is greater than 5 mm, greater than 10 mm, greater than 20 mm, greater than 30 mm, greater than 50 mm, greater than 100 mm, greater than 200 mm, greater than 500 mm, or even larger. The area of the substrate **502** occupied by the set **511** can be of any suitable or desired, symmetric or asymmetric, regular or irregular shape (e.g., polygonal, circular, elliptical, oval, ring, annulus, shape of an eyewear lens or of a visor or faceplate of a helmet, and so forth). In some examples the light-emitting elements **513** of the set **511** can occupy a non-zero fraction of the area **506** of the substrate **502** occupied by the set **511** that is less than 25%, less than 10%, less than 5%, less than 2%, less than 1%, less than 0.5%, less than 0.2%, less than 0.1%, less than 0.05%, less than 0.02%, less than 0.01%, less than 0.005%, less than 0.002%, less than 0.001%, or even smaller.

[0036] In some examples the electrically conductive traces **525** can be less than 100 μm wide, less than 50 μm , less than 30 μm , less than 20 μm , less than 10 μm , less than 8 μm , less than 5 μm , or even smaller. In some examples the electrically conductive traces **525** occupy less than 25% of the areal extent of the set of electrically conductive traces, or less than 10%, less than 5%, less than 2%, less than 1%, less than 0.5%, less than 0.2%, less than 0.1%, less than 0.05%, less than 0.02%, less than 0.01%, less than 0.005%, less than 0.002%, less than 0.001%, or even smaller. In a manner similar to that of the light-emitting element **513**, those sizes or areal fractions of the electrically conductive traces **525** enable the scene **600** to be viewed along a line of sight **598** that passes through the traces **525**, for the traces **525** not to substantially interfere with visual observation of the scene **600** by the naked eye of a human observer, for the traces to be only negligibly visible to the naked eye of a human observer, or to resemble hair or dust on the substrate **502**. In some examples the set of electrically conductive traces can occupy an area of the substrate **502** having a smallest transverse dimension that is greater than 5 mm, greater than 10 mm, greater than 20 mm, greater than 30 mm, greater than 50 mm, greater than 100 mm, greater than 200 mm, greater than 500 mm, or even larger. In some examples the traces might extend beyond the areal extent **506** of the set **511**, for example, in arrangements wherein the areal extent **506** of the set **511** covers only a portion of the substrate **502** but some of the traces **525** extend to one or more edges of the substrate **502** to enable one or more electrical connections to, e.g., a control circuit or a power source.

[0037] In some examples (e.g., illustrated schematically in FIG. 6) the substrate **502** can be mounted on or attached to, or form a portion of, an object worn or carried by the human observer (e.g., eyewear **530** or a lens, eyepiece, mask, faceplate, visor, headset, helmet, and so forth). In some examples (e.g., illustrated schematically in FIGS. 7 and 12) the substrate **502** can be mounted on or attached to, or form a portion of, a transparent window **544** (with light source **500RW**) or a reflective mirror **542** (with light source **500M**) of a vehicle **540**, building, or other structure or object. In some examples, the entire substrate **502** can be planar (e.g., a planar window, mirror, or screen). In some examples the substrate **502** can include two or more flat portions that are not coplanar (e.g., portions of a faceted window or a foldable display). In some examples one or more portions of the substrate **502**, or all of the substrate **502**, can be curved (e.g., a curved windshield or rear window of a vehicle, eyewear lenses, or a faceplate or visor for a headset or helmet). In some examples the substrate **502** can be a window, lens, eyepiece, mirror, display, or other structure; in some examples the substrate **502** can be a distinct structure (e.g., a flexible, transparent polymer layer) that is attached to a window, lens, eyepiece, display, mirror, or other rigid structure.

[0038] In some examples the light source **500** can be arranged so that all of the light-emitting elements **513** of the set **511** emit output light **581** to propagate away from the same surface of the substrate **502**. That is necessarily the case when the substrate **502** is reflective and the light source **500** is arranged in the reflective geometry of FIG. 1B; all of the light-emitting elements **513** would emit output light **581** to propagate generally toward the human observer **599**. With a transparent substrate **502** and the light source **500** arranged in the transmissive geometry of FIG. 1A, all of the light-

emitting elements **513** can be directed either forward (i.e., toward the scene **600** and away from the human observer) or backward (i.e., toward the human observer and away from the scene **600**). In some other examples the light source **500** can be arranged so that some of the light-emitting elements **513** of the set **511** emit output light **581** in the forward direction and other light-emitting elements **513** of the set **511** emit output light **581** in the backward direction (as already noted above).

[0039] In some examples the light source **500** can be mounted or positioned relative to the human observer so that at least a portion of output light **581** emitted by the set **511** is directed at the human observer (i.e., is directed in the backward direction, as discussed above). Output light **581** emitted by the set **511** in the backward direction can be intended to be seen by the human observer (e.g., for presenting alphanumeric, symbolic, or image content for the human observer to see), or can be used for illuminating the human observer (e.g., for tracking eye or head movements or positions, facial or retinal detection or identification, or other biometric detection or measurement). In some instances output light **581** used for illuminating the human observer can be infrared light, so that the human observer is unaware of and undisturbed by the illuminating light; note however that visible light could be used for such illumination of the human observer. In some instances the output light **581** emitted by the light source **500** can be visible light used to display alphanumeric, symbolic, or image content on the substrate **502** to be seen by the human observer. Such displayed content formed by the light source **500** can be superimposed on the scene **600** using the display **699** (e.g., as part of an AR or VR visualization system). In some instances the set **511** can include light-emitting elements **513** that serve both purposes. For example, one subset of infrared light-emitting elements **513** can be arranged for illuminating the observer's eyes or face, and another subset of visible light-emitting elements **513** can be arranged for displaying content to the observer. Such a dual-purpose arrangement could be positioned, e.g., on a vehicle windshield or rear-view mirror **542** (e.g., light source **500M** as in FIG. 7) or on eyewear **530** (e.g., light source **500** as in FIG. 6) or a headset, and could be employed, e.g., to monitor the driver's or wearer's attention (through tracking eye or head movements) and simultaneously to display information to the driver or wearer (such as speed, heading, position, outside temperature, fuel status, and so forth) or superimpose markers or indicators onto the scene **600** observed by the driver or wearer. Note that the driver or wearer is the human observer **599** in such examples.

[0040] Forward-directed light-emitting elements **513** can be arranged to serve analogous purposes. For example, forward-directed infrared or visible light-emitting elements **513** can be employed for illuminating people or objects in the scene **600**. In some instances visible light might be employed for general illumination of the scene **600**; in other instances infrared or visible illumination can be employed for tracking, range-finding, identification or recognition, or other characterizations or measurements of people or objects within the scene **600**. In some instances forward-directed visible light-emitting elements **513** can be employed for displaying alphanumeric, symbolic, or image content to other observers who are located within the scene **600**. In some instances forward-directed light emitting elements **513** of both types can be employed together, e.g., on a rear

window 544 of a vehicle 540 (e.g., light source 500RW as in FIG. 7) for sensing when a following vehicle follows too closely and to display a message politely suggesting to the driver of the following vehicle that he/she should back off, all without interfering with the driver's (i.e., observer's 599) ability to see through the rear window 544. In the example of FIG. 12, the light source 500RW is shown arranged as a brake light on the rear window 544 of the vehicle 540 (e.g., a so-called center high-mount stop light, or CHMSL), with red output light directed away from the observer/driver 599 and away from the vehicle along the sight line 598.

[0041] The light-emitting elements 513 can be formed on, retained on, attached to, or adhered to the substrate 502 (rigid or flexible) in any suitable way, for example by adhesive, welding or soldering, retention by a covering layer or structure, in situ fabrication or deposition on the substrate 502, transfer onto the substrate 502 from another substrate or carrier, and so forth. Similarly, the electrically conductive traces 525 can be formed on, retained on, attached to, or adhered to the substrate 502 in any suitable way (before, after, or concurrent with the light-emitting elements 513), for example by adhesive, welding or soldering, retention by a covering layer or structure, in situ fabrication or deposition on the substrate 502, transfer onto the substrate 502 from another substrate or carrier, and so forth. In some examples the light-emitting elements 513 and the electrically conductive traces 525 can be formed on, retained on, attached to, or adhered to the substrate 502 using distinct corresponding process sequences performed one after the other. In some other examples the light-emitting elements 513 and the electrically conductive traces 525 can both be formed on, retained on, attached to, or adhered to the substrate 502 in a single, integrated process sequence. In some examples each light-emitting element 513 can be transferred individually from a carrier (e.g., a substrate on which it was formed, or to which it was transferred after its formation) to the substrate 502 for attachment. In some other examples groups of multiple light-emitting elements 513 (comprising some or all of the set 511) can be transferred from such a carrier to the substrate 502 for attachment (e.g., by so-called mass transfer). Those examples and others included in the process diagram of FIG. 9.

[0042] Specific examples are illustrated schematically in FIGS. 8A and 8B. The light-emitting elements 513 are disposed on a flexible polymer layer 502F, in many instances along with electrically conductive traces 525. In some instances the light-emitting elements 513 can be formed directly on the polymer layer 502F, typically with the layer 502F supported by an underlying rigid support; in some other instances the light-emitting elements 513 can be transferred to and attached to the polymer layer 502F. A layer of adhesive 517 can be applied to one or both of the flexible polymer layer 502F and a rigid layer 502R; the adhesive 517 is applied to the polymer layer 502F and the light-emitting elements 513 thereon in the examples shown in FIGS. 8A and 8B. The rigid substrate 502R can be flat (e.g., as in FIG. 8A) or curved (e.g., as in FIG. 8B) or otherwise non-planar (e.g., faceted). The flexible layer 502F and the rigid layer 502R are brought together (indicated by the large hollow arrow in FIGS. 8A and 8B) to form a laminated structure that is transparent for visible light, with the light-emitting elements 513 and the adhesive 517 between the layers 502F and 502R. In some examples the adhesive 517 can be cured or cross-linked to effect adhesion

of the layers 502F and 502R. Any suitable adhesive can be employed; in some examples the adhesive includes one or more silicones, one or more epoxies, or one or more clear polyimides. In this example, before attachment of the polymer layer 502F to the rigid layer 502R, the polymer layer 502F corresponds to the substrate 502 described above. After attachment of the polymer layer 502F to the rigid layer 502R with the light-emitting elements 513 between them, the laminated structure formed by that attachment corresponds to the substrate 502 described above.

[0043] If the rigid layer 502R is curved (e.g., if the rigid layer 502R forms a curved lens or a curved visor or faceplate of a helmet or headset), the flexible polymer layer 502F can be deformed to conform to the curved shape of the rigid layer 502R. In that way it is possible to attach the light-emitting elements 513 to a rigid layer 502R of any desired or necessary shape while enabling fabrication of those light-emitting elements 513 on a planar wafer or substrate (typically required for using many semiconductor fabrication techniques used to form LEDs). The adhesive 517 adheres the layers 502F and 502R together and at least partly encapsulates the light-emitting elements 513 between them. When conforming the flexible polymer layer 502F to the shape of the rigid layer 502R (whether flat, curved, or otherwise nonplanar), attachment of the layers 502F/502R together can include taking steps to prevent, avoid, or eliminate bubbles or voids between the rigid layer 502R and the polymer layer 502F.

[0044] In some examples the flexible polymer layer 502F can remain in its flexible or deformable state after attachment to the rigid layer 502R. In some examples the polymer layer 502F can be cured or cross-linked after being positioned on and conformed to the rigid substrate 502R. In some of those latter examples the cured or cross-linked polymer layer 502F can be rigid after curing or cross-linking; in some other examples the cured or cross-linked polymer can remain somewhat flexible, pliable, or compressible. In some examples the polymer layer 502F can self-adhere to the rigid layer 502R, i.e., acting as its own adhesive. In some examples the flexible polymer layer 502F can include one or more materials among clear polyimide (PI), polyethylene naphthalate (PEN), polyethylene terephthalate (PET), or other flexible or curable transparent polymer. In some examples the rigid layer 502R can include one or more materials among silica, optical glasses, polycarbonate, polymethylmethacrylate (PMMA), other rigid transparent polymers, or other rigid transparent materials.

[0045] In any application (including examples described herein) wherein the light-emitting elements 513 are employed for any sort of sensing or detection, a sensor or detector or imaging device of any suitable type (generically labelled 585 in FIG. 10B) can be employed to receive light 581 emitted by the illuminating light-emitting elements 513 and then reflected or scattered from the human observer or from people or objects within the scene 600. In the example of FIG. 6, the detector or sensor 585 might be located on the frame of the eyewear 530; in the example of FIG. 7, the detector or sensor 585 might be located on a frame around the rearview mirror 542 or around the rear window 544; any suitable location can be employed. A computer or other suitable processing system or control circuit (generically labelled 553 in FIG. 10B) typically would be employed to interpret the acquired signals or images according to a suitable algorithm or protocol. In the AR/VR example, an

infrared image sensor **585** can receive infrared light, emitted as output light **581** by the light-emitting elements **513** on the AR/VR eyewear or headset **530**, that is scattered or reflected from the user's eyes. A computer or processor or control circuit **553** operatively coupled to the sensor or detector **585** can estimate or determine a direction of the user's gaze using a suitable eye-tracking algorithm, and the AR/VR system can be further programmed to alter the AR/VR imagery presented to the user using the display **699** (e.g., highlighting a person or object in the scene **600**, displaying a reticle or crosshair superimposed on the scene **600**, displaying data or information superimposed on the scene **600** pertaining to a person or object in the scene **600**, and so on).

[0046] As noted above, in some examples of a transmissive geometry the output light **581** from the set **511** of light-emitting elements **513** is directed away from the observer **599**, for sensing portions of the scene **600**, for illuminating the scene **600**, or to be seen by another observer located within the scene **600**. One example of that latter scenario would be the light source **500RW** acting as a high-mount brake light on the rear window **544** of the vehicle **540**, schematically illustrated in FIG. 12. Using a low-density light source **500RW** for the high-mount brake light can enable the observer/driver **599** to see through the brake light along the sight line **598** when looking into the rearview mirror **542**; the brake light need not obstruct a portion of the rear window **544**, as is the case with conventional high-mount brake lights. Note that in the arrangement of FIG. 12, although the visible output light is propagating backward relative to the vehicle, the arrangement of the light source **500RW** would nevertheless be considered forward-directed since its output light propagates away from the observer **599** along the sight line **598**. Although the visible output light (typically red light for a vehicle brake light) is directed generally away from the driver/observer **599**, often some of that light can propagate, unwanted, toward the observer **599**. In some examples, including the brake light arrangement of FIG. 12, that undesirable propagation of visible output light toward the observer **599** can interfere with observation of the scene **600** through the substrate **502** (e.g., through the vehicle rear window **544**).

[0047] Accordingly, in some examples (e.g., as illustrated schematically in FIG. 13) a visible-emitting low-density light source **500** can include an optical attenuator structure **560**. The light-emitting elements **513** on the transparent substrate **502** are arranged to generate and emit visible output light to propagate away from a first surface of the substrate **502** primarily in a first direction (toward the scene **600** and away from the observer **599**). The optical attenuator structure **560** can be positioned on or facing a second surface of the substrate **502** opposite the first surface, between the light-emitting elements **513** and the observer **599**. In some examples the optical attenuator structure **560** can substantially continuously spanning the entire areal extent **506** of the set **511** of the light-emitting elements **513**. The optical attenuator structure **560** can be positioned and arranged to attenuate or block transmission any of the visible output light that propagates toward the observer **599**, while still transmitting enough visible light so that the observer **599** can observe the scene **600** along the sight line **598** through the optical attenuator structure **560** and through the set **511** of the light-emitting elements **513**. In the example of FIG. 13 the optical attenuator structure is shown positioned against the substrate **502**. In some of those examples the optical

attenuator structure **560** can be formed on the surface of the substrate **502**; in some other of those examples the optical attenuator structure **560** can be formed on another substrate, or as a self-supporting structure, and then assembled with the substrate **502**. In some examples the optical attenuator structure need not be positioned against the substrate **502**, but can be a separate structure spaced apart from the substrate **502** along the sight line **598**.

[0048] In some examples the optical attenuator structure **560** can be arranged to selectively attenuate or block transmission of output light at a vacuum wavelength to, emitted by the light-emitting elements **513**, relative to transmission of light at some other visible wavelengths. Such a wavelength-selective optical attenuator structure **560** can be arranged as, e.g., a short-pass, long-pass, or notch optical filter that selectively attenuates or blocks transmission of output light at the vacuum wavelength **2**. Such a wavelength-selective optical attenuator structure **560** would still enable the observer to see through it, although the color of the scene **600** typically would appear different due to the missing wavelengths blocked by the attenuator structure **560**.

[0049] The wavelength-selective optical attenuator structure **560** can be of any suitable type or arrangement. In some instances such wavelength-selective optical attenuators can be of an absorptive type, e.g., optical glass or other transparent host material doped with one or more chemical species (e.g., dyes or ions) that absorb light selectively at the wavelength(s) emitted by the light-emitting elements **513** while transmitting other visible wavelengths. Absorptive optical filters (e.g., colored glass filters) have the desirable feature that their absorption/transmission spectra typically do not vary with incidence angle; they have the disadvantage the absorbed output light is lost, decreasing the overall efficiency of the light source **500**. Examples abound of commercially available optical filters exhibiting a variety of absorption spectra; a suitable example could be selected based on the wavelength emitted by the light-emitting elements **513**.

[0050] In some instances a reflective wavelength-selective optical attenuator structure **560** can be employed, typically in the form of a suitably designed multilayer dielectric coating. Such coatings can be designed as short-pass, long-pass, or notch filters, often with relatively sharp transitions from transmission to reflection as a function of wavelength. Again, such reflective filters are readily available commercially at virtually any desired design wavelength and, in case of a notch filter, any desired width of a blocked band of wavelengths. When such reflective filters are employed, the output light propagating toward the observer **599** that is blocked is also reflected back toward the scene **600**, increasing the overall efficiency of the light source **500**. A disadvantage of multilayer dielectric filters is the dependence of their transmission/reflection spectra on angle of incidence. With increasing angle of incidence, the characteristic wavelength of a multilayer filter (cut-off wavelength for a short- or long-pass filter, center wavelength for a notch filter) shifts to shorter wavelengths. Such angle dependence can make such filters unsuitable in certain instances where a large range of observation angles might be needed. However, in many instances the observation angle (i.e., angle between the filter normal vector and the sight line **598**) might be narrow enough that a multilayer dielectric filter can be suitably employed as the attenuator structure **560**. In par-

ticular, in the example of FIG. 12, the relatively small sizes of the rearview mirror 542 and the light source 500RW (i.e., the brake light) result in a relative narrow range of observation angles. A multilayer dielectric stack filter (typically notch or short-pass) can be readily designed that blocks the red light emitted by the brake light from propagating to the driver/observer 599 by reflection from the rearview mirror 542, over a range of angles that encompass all sight lines from the observer/driver 599, reflected from the rearview mirror 542, and out through the rear window 544 through the light source 500RW.

[0051] Instead of, or in addition to, a wavelength-selective optical attenuator structure, in some examples the optical attenuator 560 can be arranged as a partially reflecting and partially transmitting reflector across the visible spectrum (e.g., as a reflective neutral density filter). Such an attenuator structure would only partly attenuate output light 581 from reaching the observer 599, and would transmit only a portion of light from the scene 600 to reach the observer 599, but can nevertheless improve the ability of the observer 599 to see the scene 600 in spite of some of the output light 581 reaching the observer 599. In some examples, the optical attenuator structure 560 can be greater than 10% reflective, greater than 20% reflective, greater than 30% reflective, greater than 40% reflective, greater than 50% reflective, less than 50% reflective, less than 60% reflective, less than 70% reflective, less than 80% reflective, or less than 90% reflective.

[0052] In addition to the preceding, the following example embodiments fall within the scope of the present disclosure or appended claims. Any given Example below that refers to one or some or all preceding Examples shall be understood to refer to only those preceding Examples with which the given Example is not inconsistent, and to exclude those preceding Examples with which the given Example is inconsistent.

[0053] Example 1. A light source comprising: (a) a substrate that is transparent or reflective for visible light; and (b) a set of multiple light-emitting elements positioned on or within the substrate, each light-emitting element comprising one or more inorganic microLEDs that are arranged to generate and emit output light to propagate out-of-plane relative to a corresponding localized area of the substrate around that light-emitting element, (c) each light-emitting element of the set being sufficiently small in at least one transverse dimension, and the light-emitting elements occupying a sufficiently small fraction of an areal extent of the set, so as to enable visual observation of a scene through or reflected by the substrate along a sight line that passes through the set of light-emitting elements.

[0054] Example 2. The light source of Example 1, the substrate being transparent for visible light, and the light source being arranged so as to enable visual observation of the scene through the substrate along the sight line that passes through the set of light-emitting elements.

[0055] Example 3. The light source of any one of Examples 1 or 2, the substrate being reflective for visible light, and the light source being arranged so as to enable visual observation of the scene reflected by the substrate along the sight line that passes through the set of light-emitting elements.

[0056] Example 4. The light source of any preceding Example, each light-emitting element of the set being arranged and connected so as to be operable independently

of at least one other light-emitting element of the set, or independently of all other light-emitting elements of the set.

[0057] Example 5. The light source of any preceding Example, one or more or all of the microLEDs being arranged to generate and emit visible output light.

[0058] Example 6. The light source of any preceding Example, one or more or all of the microLEDs comprising UV- or visible-emitting, direct-emitting or phosphor-converted, semiconductor microLEDs, each microLED including one or more materials among III-V, II-VI, or Group IV semiconductor materials.

[0059] Example 7. The light source of any preceding Example, one or more or all of the microLEDs being arranged to generate and emit non-visible output light.

[0060] Example 8. The light source of any preceding Example, one or more or all of the microLEDs comprising infrared-emitting, direct-emitting or phosphor-converted, semiconductor microLEDs, each microLED including one or more materials among III-V, II-VI, or Group IV semiconductor materials.

[0061] Example 9. The light source of any preceding Example, one or more or all of the light-emitting elements of the set each including only a single microLED.

[0062] Example 10. The light source of any preceding Example, one or more or all of the light-emitting elements of the set each including multiple microLEDs, each microLED of a given light-emitting element being arranged and connected so as to be operable independently of at least one other microLED of that light-emitting element, or independently of all other light-emitting elements of that light-emitting element.

[0063] Example 11. The light source of any preceding Example, one or more or all of the microLEDs including a wavelength-converting structure.

[0064] Example 12. The light source of any preceding Example, each light-emitting element having a largest transverse dimension that is less than 200 μm , less than 100 μm , less than 50 μm , less than 30 μm , less than 20 μm , less than 10 μm , less than 8 μm , less than 5 μm , or even smaller.

[0065] Example 13. The light source of any preceding Example, the light-emitting elements occupying less than 25% of the areal extent of the set, less than 10%, less than 5%, less than 2%, less than 1%, less than 0.5%, less than 0.2%, less than 0.1%, less than 0.05%, less than 0.02%, less than 0.01%, less than 0.005%, less than 0.002%, less than 0.001%, or even smaller.

[0066] Example 14. The light source of any preceding Example, the set of light-emitting elements occupying an area of the substrate having a smallest transverse dimension that is greater than 5 mm, greater than 10 mm, greater than 20 mm, greater than 30 mm, greater than 50 mm, greater than 100 mm, greater than 200 mm, greater than 500 mm, or even larger.

[0067] Example 15. The light source of any one of Examples 1 through 14, the entire substrate being planar.

[0068] Example 16. The light source of any one of Examples 1 through 14, the substrate including one or more areal regions that are curved, or two or more planar areal regions that are not coplanar with respect to one another.

[0069] Example 17. The light source of any one of Examples 1 through 16, the substrate being flexible.

[0070] Example 18. The light source of any one of Examples 1 through 16, the substrate being rigid.

[0071] Example 19. The light source of any preceding Example, each light-emitting element being sufficiently small in at least one transverse dimension, and the light-emitting elements of the set being spaced sufficiently far apart, so that, with the light-emitting elements in an off state or emitting only non-visible light, the set does not substantially interfere with visual observation of the scene through or reflected by the substrate by a naked eye of a human observer along the sight line that passes through the set.

[0072] Example 20. The light source of any preceding Example, each light-emitting element being sufficiently small in at least one transverse dimension, and the light-emitting elements of the set being spaced sufficiently far apart, so that to a naked eye of a human observer the set is only negligibly visible when in an off state or when emitting only non-visible light.

[0073] Example 21. The light source of any preceding Example, each light-emitting element being sufficiently small in at least one transverse dimension, and the light-emitting elements of the set being spaced sufficiently far apart, so that to a naked eye of a human observer the set resembles dust on the substrate when in an off state or when emitting only non-visible light.

[0074] Example 22. The light source of any preceding Example further comprising multiple electrically conductive traces on or within the substrate that are arranged and connected for providing electrical drive current to the light-emitting elements of the set, the traces being sufficiently transparent, sufficiently narrow, or spaced sufficiently far apart so as to enable visual observation of the scene through or reflected by the substrate along the sight line that passes through the set of light-emitting elements and the set of electrically conductive traces.

[0075] Example 23. An apparatus comprising: (a) a substrate that is transparent for visible light; and (b) a set of multiple electrically conductive traces on or within the substrate, the electrically conductive traces being sufficiently transparent, or being less than 200 μm wide and occupying less than 25% of an areal extent of the set, so as to enable visual observation of a scene through the substrate along a sight line that passes through the set of electrically conductive traces.

[0076] Example 24. An apparatus comprising: (a) a substrate that is reflective for visible light; and (b) a set of multiple electrically conductive traces on or within the substrate, the electrically conductive traces being sufficiently transparent, or being less than 200 μm wide and occupying less than 25% of an areal extent of the set, so as to enable visual observation of a scene reflected by the substrate along a sight line that passes through the set of electrically conductive traces.

[0077] Example 25. The apparatus of any one of Examples 22 through 24, the set of electrically conductive traces including one or more materials among indium tin oxide (ITO), indium zinc oxide (IZO), or one or more other transparent conductive oxides (TCOs).

[0078] Example 26. The apparatus of any one of Examples 22 through 25, the set of electrically conductive traces including one or more materials among aluminum, silver, gold, or one or more other metals or metallic alloys.

[0079] Example 27. The apparatus of any one of Examples 22 through 26, the electrically conductive traces being less

than 200 μm , less than 100 μm , less than 50 μm , less than 30 μm , less than 20 μm , less than 10 μm , less than 8 μm , less than 5 μm , or even smaller.

[0080] Example 28. The apparatus of any one of Examples 22 through 27, the electrically conductive traces occupying less than 25% of the areal extent of the set, less than 10%, less than 5%, less than 2%, less than 1%, less than 0.5%, less than 0.2%, less than 0.1%, less than 0.05%, less than 0.02%, less than 0.01%, less than 0.005%, less than 0.002%, less than 0.001%, or even smaller.

[0081] Example 29. The apparatus of any one of Examples 22 through 28, the set of electrically conductive traces occupying an area of the substrate having a smallest transverse dimension that is greater than 5 mm, greater than 10 mm, greater than 20 mm, greater than 30 mm, greater than 50 mm, greater than 100 mm, greater than 200 mm, greater than 500 mm, or even larger.

[0082] Example 30. The apparatus of any one of Examples 22 through 29, the electrically conductive traces being sufficiently transparent, or sufficiently narrow and spaced sufficiently far apart, so that the set does not substantially interfere with visual observation of the scene through or reflected by the substrate by a naked eye of a human observer along the sight line that passes through the set.

[0083] Example 31. The apparatus of any one of Examples 22 through 30, the electrically conductive traces being sufficiently transparent, or sufficiently narrow and spaced sufficiently far apart, so that to a naked eye of a human observer the set is only negligibly visible.

[0084] Example 32. The apparatus of any one of Examples 22 through 31, the electrically conductive traces being sufficiently transparent, or sufficiently narrow and spaced sufficiently far apart, so that to a naked eye of a human observer the set resembles hair or dust on the substrate.

[0085] Example 33. The light source of any preceding Example, the substrate comprising a rigid layer and a polymer layer attached to the rigid layer, the rigid layer and the polymer layer forming a laminated structure that is transparent for visible light, the set of multiple light-emitting elements being positioned between the rigid layer and the polymer layer.

[0086] Example 34. A light source comprising: (a) a rigid layer; (b) a polymer layer attached to the rigid layer, the rigid layer and the polymer layer forming a laminated structure that is transparent for visible light; and (c) a set of multiple light-emitting elements positioned between the rigid layer and the polymer layer, each light-emitting element comprising one or more inorganic microLEDs that are arranged to generate and emit output light to propagate out-of-plane relative to a corresponding localized area of the laminated structure surrounding that light-emitting element, (d) each light-emitting element of the set being sufficiently small in at least one transverse dimension, and the light-emitting elements occupying a sufficiently small fraction of an areal extent of the set, so as to enable visual observation of a scene through the laminated structure along a sight line that passes through the set of light-emitting elements.

[0087] Example 35. The light source of any one of Examples 33 or 34 further comprising an adhesive layer between the rigid layer and the polymer layer, the adhesive layer adhering together the rigid layer and the polymer layer and at least partly encapsulating the set of light-emitting elements, the adhesive layer being transparent for visible light.

[0088] Example 36. The light source of any one of Examples 33 or 34, the polymer layer self-adhering to the rigid layer and at least partially encapsulating the set of light-emitting elements.

[0089] Example 37. The light source of any one of Examples 33 through 36, the entire rigid layer being planar.

[0090] Example 38. The light source of any one of Examples 33 through 36, the rigid layer including one or more areal regions that are curved, or two or more planar areal regions that are not coplanar with respect to one another.

[0091] Example 39. The light source of any one of Examples 33 through 38, the polymer layer being flexible.

[0092] Example 40. The light source of any one of Examples 33 through 38, the polymer layer comprising a rigid layer of cured or cross-linked polymer material.

[0093] Example 41. The light source of any one of the preceding Examples, (i) the polymer layer including one or more polymeric materials among clear polyimide (PI), polyethylene naphthalate (PEN), polyethylene terephthalate (PET), or other flexible or curable or cross-linked transparent polymer, or (ii) the rigid layer including one or more materials among silica, optical glasses, polycarbonate, polymethylmethacrylate (PMMA), or other rigid transparent polymers.

[0094] Example 42. A method for making the light source of any one of the preceding Examples, the method comprising: (A) providing the set of multiple light-emitting elements disposed on the polymer layer; and (B) attaching the rigid layer to the polymer layer with the set of light-emitting elements therebetween, the rigid layer and the polymer layer forming the laminated structure, (C) wherein, before attachment to the rigid layer, the polymer layer is flexible.

[0095] Example 43. A method comprising: (A) providing a set of multiple light-emitting elements disposed on a flexible polymer layer, each light-emitting element comprising one or more inorganic microLEDs that are arranged to generate and emit output light to propagate out-of-plane relative to a corresponding localized area of the polymer layer surrounding that light-emitting element; and (B) attaching a rigid layer to the polymer layer with the set of light-emitting elements therebetween, the rigid layer and the polymer layer forming a laminated structure that is transparent for visible light, each light-emitting element of the set being sufficiently small in at least one transverse dimension, and the light-emitting elements occupying a sufficiently small fraction of an areal extent of the set, so as to enable visual observation of a scene through the laminated structure along a sight line that passes through the set of light-emitting elements.

[0096] Example 44. The method of any one of Examples 42 or 43 wherein providing the set of light-emitting elements includes transferring individually each light-emitting element from a carrier onto the polymer layer.

[0097] Example 45. The method of any one of Examples 42 or 43 wherein providing the set of light-emitting elements includes transferring simultaneously groups of multiple light-emitting elements from a carrier onto the polymer layer.

[0098] Example 46. The method of any one of Examples 42 through 45, each light-emitting element of the set including multiple microLEDs, the method further comprising arranging and connecting each microLED of a given light-

emitting element so as to be operable independently of at least one other light emitter of that light-emitting element.

[0099] Example 47. The method of any one of Examples 42 through 46, further comprising arranging and connecting each light-emitting element of the set so as to be operable independently of at least one other light-emitting element of the set.

[0100] Example 48. The method of any one of Examples 42 through 47 further comprising forming on the polymer layer multiple electrically conductive traces that are arranged and connected for providing electrical drive current to the light-emitting elements of the set, the traces being sufficiently transparent, sufficiently narrow, or spaced sufficiently far apart so as to enable visual observation of the scene through the laminated structure along the sight line that passes through the set of light-emitting elements.

[0101] Example 49. The method of any one of Examples 42 through 48 further comprising applying an adhesive layer between the rigid layer and the polymer layer, the adhesive layer adhering together the rigid layer and the polymer layer and at least partly encapsulating the set of light-emitting elements, the adhesive layer being transparent for visible light.

[0102] Example 50. The method of any one of Examples 42 through 48, the polymer layer self-adhering to the rigid layer and at least partially encapsulating the set of light-emitting elements.

[0103] Example 51. The method of any one of Examples 42 through 50 wherein attaching the rigid layer to the polymer layer includes preventing, avoiding, or eliminating bubbles or voids between the rigid layer and the polymer layer.

[0104] Example 52. The method of any one of Examples 42 through 51, the polymer layer remaining flexible after attachment to the rigid layer.

[0105] Example 53. The method of any one of Examples 42 through 52 wherein attaching the rigid layer to the polymer layer includes curing or cross-linking the polymer layer.

[0106] Example 54. The method of Example 53, the polymer layer being rigid after curing or cross-linking.

[0107] Example 55. The method of any one of Examples 53 or 54, the polymer layer self-adhering to the rigid layer after curing or cross-linking.

[0108] Example 56. A method for operating the light source of any one of Examples 1 through 41, the method comprising, with the substrate positioned in a sight line along which a scene is observed through the substrate by an observer, operating the set of multiple light-emitting elements to emit the output light.

[0109] Example 57. The method of Example 56, some or all of the output light being directed toward the observer.

[0110] Example 58. The method of any one of Examples 56 or 57, some or all of the output light being directed toward the scene.

[0111] Example 59. A method for operating the light source of any one of Examples 1 through 41, the method comprising, with the substrate positioned in a sight line along which a scene is observed reflected by the substrate by an observer, operating the set of multiple light-emitting elements to emit the output light.

[0112] Example 60. A method for operating the apparatus of any one of Examples 22 through 41, method comprising, with the substrate positioned in a sight line along which a

scene is observed through the substrate by an observer, delivering one or more electrical signals or drive currents through one or more of the electrically conductive traces.

[0113] Example 61. A wearable optical assembly incorporating the light source of any one of Examples 1 through 41, the optical assembly being structurally configured so as to position the substrate in the sight line of a user wearing the optical assembly, some or all the light-emitting elements being arranged to generate and emit infrared output light, the optical assembly being structurally configured so that the output light propagates toward an eye or face of the user wearing the optical assembly.

[0114] Example 62. A wearable optical assembly comprising: (a) an optical element (such as a substrate) that is transparent for visible light, the optical assembly being structurally configured so as to position the optical element in a sight line of a user wearing the optical assembly; and (b) a set of multiple light-emitting elements positioned on or within the optical element, each light-emitting element comprising one or more inorganic microLEDs that are arranged to generate and emit infrared output light to propagate out-of-plane relative to a corresponding localized area of the optical element around that light-emitting element, the optical assembly being structurally configured so that the output light propagates toward an eye or face of the user wearing the optical assembly, (c) each light-emitting element of the set being sufficiently small in at least one transverse dimension, and the light-emitting elements occupying a sufficiently small fraction of an areal extent of the set, so as to enable visual observation of a scene, by the user wearing the optical assembly, through the optical element along the sight line of the user, the sight line passing through the set of light-emitting elements.

[0115] Example 63. The wearable optical assembly of any one of Examples 61 or 62 further comprising: one or more infrared detectors or infrared image sensors, the optical assembly being structurally configured so as to position the one or more infrared detectors or infrared image sensors for receiving portions of the infrared output light scattered or reflected from an eye or face of the user wearing the optical assembly, the one or more infrared detectors or infrared image sensors being arranged for generating corresponding electronic signals indicative of the received scattered or reflected light; and one or more electronic circuits or electronic processors structured, connected, or programmed (i) for receiving one or more of the electronic signals generated by one or more of the infrared detectors or infrared image sensors, and (ii) for eye tracking, facial recognition, or optical biometric measurement based on one or more of the electronic signals received from the one or more infrared detectors or infrared image sensors.

[0116] Example 64. The wearable optical assembly of any one of Examples 61 through 63, the wearable optical assembly being arranged as eyeglasses, eyewear, goggles, a headset, a helmet, or a head-mounted device, the optical element forming at least a portion of a window, lens, eyepiece, display screen, faceplate, or visor of the wearable optical assembly.

[0117] Example 65. A method comprising, with a user wearing the wearable optical assembly of any one of Examples 61 through 64, operating the set of multiple light-emitting elements to generate and emit the infrared output light to propagate toward an eye or face of the user wearing the optical assembly.

[0118] Example 66. The method of Example 65 further comprising: receiving at one or more infrared detectors or infrared image sensors portions of the infrared output light scattered or reflected from an eye or face of the user wearing the optical assembly, and generating corresponding electronic signals indicative of the received scattered or reflected light; and based at least in part on one or more of the electronic signals generated by and received from the one or more infrared detectors or infrared image sensors, using one or more electronic circuits or electronic processors structured, connected, or programmed therefor, (i) tracking movement of one or both eyes of the user wearing the optical assembly, (ii) recognizing a face of the user wearing the optical assembly, or (iii) making an optical biometric measurement of the user wearing the optical assembly.

[0119] Example 67. The method of Example 66 further comprising, based at least in part on one or more of the electronic signals generated by and received from the one or more infrared detectors or infrared image sensors, using one or more electronic circuits or electronic processors structured, connected, or programmed therefor, (i) tracking movement of one or both eyes of the user wearing the optical assembly, and (ii) based at least in part on the tracked eye movement, taking one or more actions among illuminating a selected portion of the scene, highlighting a selected portion of the scene, or presenting to the wearer of the optical assembly alphanumeric or graphic information superimposed on the scene in the sight line of the user wearing the optical assembly.

[0120] Example 68. The light source of any one of the transmissive, visible-emitting Examples among Examples 1 through 41, or made according to any one of the methods of Examples 42 through 55, further comprising an optical attenuator structure positioned on so as to block or attenuate visible light emitting by the set of light-emitting elements that propagates in a direction opposite the scene.

[0121] Example 69. A light source comprising: a substrate that is transparent for visible light; and a set of multiple light-emitting elements positioned on or within the substrate, each light-emitting element comprising one or more inorganic microLEDs that are arranged to generate and emit visible output light to propagate away from a first surface of the substrate primarily in a first direction out-of-plane relative to a corresponding localized area of the substrate around that light-emitting element, an optical attenuator structure positioned on or facing a second surface of the substrate opposite the first surface, each light-emitting element of the set being sufficiently small in at least one transverse dimension, and the light-emitting elements occupying a sufficiently small fraction of an areal extent of the set, so as to enable visual observation of a scene through the substrate along a sight line that passes through the set of light-emitting elements, the optical attenuator structure being positioned and arranged so as to (i) attenuate or block transmission therethrough of a portion of the visible output light that propagates in a second direction, opposite the first direction, and (ii) transmit sufficient visible light so as to enable visual observation of the scene along the sight line through the optical attenuator structure and through the set of light-emitting elements.

[0122] Example 70. The light source of any one of Examples 68 or 69, the one or more microLEDs emitting output light a vacuum wavelength λ , the optical attenuator structure being arranged so as to selectively attenuate or

block transmission therethrough of output light at the vacuum wavelength $2\lambda_0$, relative to transmission therethrough of light at some other visible wavelengths.

[0123] Example 71. The light source of Example 70, the optical attenuator structure being arranged as a short-pass, long-pass, or notch optical filter that attenuates or blocks transmission of output light at the vacuum wavelength $2\lambda_0$.

[0124] Example 72. The light source of any one of Examples 70 or 71: (i) the substrate being a rear window of a motor vehicle with the first direction being directed out of the vehicle, (ii) the visible output light being red output light, (iii) the optical attenuator structure being arranged as a short-pass or notch optical filter that attenuates or blocks transmission of the red output light into the vehicle, and (iv) the light source being arranged and coupled to the vehicle so as to operate as a brake light.

[0125] Example 73. The light source of any one of Examples 68 through 72, the optical attenuator structure being arranged to reflect at least a portion of the output light propagating in the second direction to propagate in the first direction.

[0126] Example 74. The light source of any one of Examples 68 through 73, the optical attenuator structure substantially continuously spanning the entire areal extent of the set of light-emitting elements.

[0127] Example 75. The light source of Example 74, the optical attenuator structure being arranged as a partially reflecting and partially transmitting reflector across the visible spectrum.

[0128] Example 76. The light source of Example 75, the optical attenuator structure being at least 10% reflective and at most 90% reflective.

[0129] Example 77. The light source of any one of Examples 68 through 76, the microLEDs comprising UV- or visible-emitting, direct-emitting or phosphor-converted, semiconductor microLEDs, the microLEDs including one or more materials among III-V, II-VI, or Group IV semiconductor materials.

[0130] Example 78. The light source of any one of Examples 68 through 77: (i) each light-emitting element having a largest transverse dimension that is less than 200 μm , (ii) the light-emitting elements occupying less than 25% of the areal extent of the set, or the set of light-emitting elements occupying an area of the substrate having a smallest transverse dimension that is greater than 5 mm.

[0131] Examples 79. The light source of any one of Examples 68 through 78, each light-emitting element of the set including only a single microLED.

[0132] Example 80. The light source of any one of Examples 68 through 78, each light-emitting element of the set including multiple microLEDs, each microLED of a given light-emitting element being arranged and connected so as to be operable independently of at least one other microLED of that light-emitting element.

[0133] Example 81. The light source of any one of Examples 68 through 80, each light-emitting element of the set being arranged and connected so as to be operable independently of at least one other light-emitting element of the set.

[0134] Example 82. The light source of any one of Examples 68 through 81 further comprising multiple electrically conductive traces on or within the substrate that are arranged and connected for providing electrical drive current to the light-emitting elements of the set, the traces being

sufficiently transparent, sufficiently narrow, or spaced sufficiently far apart so as to enable visual observation of the scene through or reflected by the substrate along the sight line that passes through the set of light-emitting elements and the set of electrically conductive traces.

[0135] Example 83. The light source of any one of Examples 68 through 82, each light-emitting element being sufficiently small in at least one transverse dimension, and the light-emitting elements of the set being spaced sufficiently far apart, so that the set does not substantially interfere with visual observation of the scene through the substrate by a naked eye of a human observer along the sight line that passes through the set.

[0136] Example 84. The light source of any one of Examples 68 through 83, each light-emitting element being sufficiently small in at least one transverse dimension, and the light-emitting elements of the set being spaced sufficiently far apart, so that to a naked eye of a human observer the set is only negligibly visible with the light-emitting elements in an off-state.

[0137] Example 85. A method employing the light source of any one of Examples 68 through 84, the method comprising, with the substrate positioned in a sight line along which a scene is observed through the substrate by an observer, operating the set of multiple light-emitting elements positioned on or within the substrate to emit visible output light.

[0138] Example 86. The method of Example 85, the one or more microLEDs emitting output light a vacuum wavelength λ_0 , the optical attenuator structure being arranged as a short-pass, long-pass, or notch optical filter that attenuates or blocks transmission of output light at the vacuum wavelength λ_0 .

[0139] Example 87. The method of Example 85: (i) the optical attenuator structure substantially continuously spanning the entire areal extent of the set of light-emitting elements, (ii) the optical attenuator structure being arranged as a partially reflecting and partially transmitting reflector across the visible spectrum, and (iii) the optical attenuator structure being at least 10% reflective and at most 90% reflective.

[0140] Example 88. The method of Example 85: (i) the substrate being a rear window of a motor vehicle with the observer being inside the vehicle and the scene being outside the vehicle, (ii) the visible output light being red output light, (iii) the optical attenuator structure being arranged as a short-pass or notch optical filter that attenuates or blocks transmission of the red output light into the vehicle, and (iv) the light source being arranged and coupled to the vehicle so as to operate as a brake light.

[0141] This disclosure is illustrative and not limiting. Further modifications will be apparent to one skilled in the art in light of this disclosure and are intended to fall within the scope of the present disclosure or appended claims. It is intended that equivalents of the disclosed example embodiments and methods, or modifications thereof, shall fall within the scope of the present disclosure or appended claims.

[0142] In the foregoing Detailed Description, various features may be grouped together in several example embodiments for the purpose of streamlining the disclosure. This method of disclosure is not to be interpreted as reflecting an intention that any claimed embodiment requires more features than are expressly recited in the corresponding claim.

Rather, as the appended claims reflect, inventive subject matter may lie in less than all features of any single disclosed example embodiment. Therefore, the present disclosure shall be construed as implicitly disclosing any embodiment having any suitable subset of one or more features-which features are shown, described, or claimed in the present application-including those subsets that may not be explicitly disclosed herein. A “suitable” subset of features includes only features that are neither incompatible nor mutually exclusive with respect to any other feature of that subset. Accordingly, the appended claims are hereby incorporated in their entirety into the Detailed Description, with each claim standing on its own as a separate disclosed embodiment. In addition, each of the appended dependent claims shall be interpreted, only for purposes of disclosure by said incorporation of the claims into the Detailed Description, as if written in multiple dependent form and dependent upon all preceding claims with which it is not inconsistent. It should be further noted that the cumulative scope of the appended claims can, but does not necessarily, encompass the whole of the subject matter disclosed in the present application.

[0143] The following interpretations shall apply for purposes of the present disclosure and appended claims. The words “comprising,” “including,” “having,” and variants thereof, wherever they appear, shall be construed as open ended terminology, with the same meaning as if a phrase such as “at least” were appended after each instance thereof, unless explicitly stated otherwise. The article “a” shall be interpreted as “one or more” unless “only one,” “a single,” or other similar limitation is stated explicitly or is implicit in the particular context; similarly, the article “the” shall be interpreted as “one or more of the” unless “only one of the,” “a single one of the,” or other similar limitation is stated explicitly or is implicit in the particular context. The conjunction “or” is to be construed inclusively unless: (i) it is explicitly stated otherwise, e.g., by use of “either . . . or,” “only one of,” or similar language; or (ii) two or more of the listed alternatives are understood or disclosed (implicitly or explicitly) to be incompatible or mutually exclusive within the particular context. In that latter case, “or” would be understood to encompass only those combinations involving non-mutually-exclusive alternatives. In one example, each of “a dog or a cat,” “one or more of a dog or a cat,” and “one or more dogs or cats” would be interpreted as one or more dogs without any cats, or one or more cats without any dogs, or one or more of each.

[0144] For purposes of the present disclosure or appended claims, when a numerical quantity is recited (with or without terms such as “about,” “about equal to,” “substantially equal to,” “greater than about,” “less than about,” and so forth), standard conventions pertaining to measurement precision, rounding error, and significant digits shall apply, unless a differing interpretation is explicitly set forth. For null quantities described by phrases such as “prevented,” “absent,” “eliminated,” “equal to zero,” “negligible,” and so forth (with or without terms such as “substantially” or “about”), each such phrase shall denote the case wherein the quantity in question has been reduced or diminished to such an extent that, for practical purposes in the context of the intended operation or use of the disclosed or claimed apparatus or method, the overall behavior or performance of the apparatus or method does not differ from that which would have

occurred had the null quantity in fact been completely removed, exactly equal to zero, or otherwise exactly nulled.

[0145] For purposes of the present disclosure and appended claims, any labelling of elements, steps, limitations, or other portions of an embodiment, example, or claim (e.g., first, second, third, etc., (a), (b), (c), etc., or (i), (ii), (iii), etc.) is only for purposes of clarity, and shall not be construed as implying any sort of ordering or precedence of the portions so labelled. If any such ordering or precedence is intended, it will be explicitly recited in the embodiment, example, or claim or, in some instances, it will be implicit or inherent based on the specific content of the embodiment, example, or claim. In the appended claims, if the provisions of 35 USC § 112(f) are desired to be invoked in an apparatus claim, then the word “means” will appear in that apparatus claim. If those provisions are desired to be invoked in a method claim, the words “a step for” will appear in that method claim. Conversely, if the words “means” or “a step for” do not appear in a claim, then the provisions of 35 USC § 112(f) are not intended to be invoked for that claim.

[0146] If any one or more disclosures are incorporated herein by reference and such incorporated disclosures conflict in part or whole with, or differ in scope from, the present disclosure, then to the extent of conflict, broader disclosure, or broader definition of terms, the present disclosure controls. If such incorporated disclosures conflict in part or whole with one another, then to the extent of conflict, the later-dated disclosure controls.

[0147] The Abstract is provided as required as an aid to those searching for specific subject matter within the patent literature. However, the Abstract is not intended to imply that any elements, features, or limitations recited therein are necessarily encompassed by any particular claim. The scope of subject matter encompassed by each claim shall be determined by the recitation of only that claim.

What is claimed is:

1. A light source comprising:

- a substrate that is transparent for visible light; and
- a set of multiple light-emitting elements positioned on or within the substrate, each light-emitting element comprising one or more inorganic microLEDs that are arranged to generate and emit visible output light to propagate away from a first surface of the substrate primarily in a first direction out-of-plane relative to a corresponding localized area of the substrate around that light-emitting element,
- an optical attenuator structure positioned on or facing a second surface of the substrate opposite the first surface,
- each light-emitting element of the set being sufficiently small in at least one transverse dimension, and the light-emitting elements occupying a sufficiently small fraction of an areal extent of the set, so as to enable visual observation of a scene through the substrate along a sight line that passes through the set of light-emitting elements,
- the optical attenuator structure being positioned and arranged so as to
- (i) attenuate or block transmission therethrough of a portion of the visible output light that propagates in a second direction, opposite the first direction, and

(ii) transmit sufficient visible light so as to enable visual observation of the scene along the sight line through the optical attenuator structure and through the set of light-emitting elements.

2. The light source of claim 1, the one or more microLEDs emitting output light a vacuum wavelength λ_0 , the optical attenuator structure being arranged so as to selectively attenuate or block transmission therethrough of output light at the vacuum wavelength λ_0 , relative to transmission therethrough of light at some other visible wavelengths.

3. The light source of claim 2, the optical attenuator structure being arranged as a short-pass, long-pass, or notch optical filter that attenuates or blocks transmission of output light at the vacuum wavelength λ_0 .

4. The light source of claim 2: (i) the substrate being a rear window of a motor vehicle with the first direction being directed out of the vehicle, (ii) the visible output light being red output light, (iii) the optical attenuator structure being arranged as a short-pass or notch optical filter that attenuates or blocks transmission of the red output light into the vehicle, and (iv) the light source being arranged and coupled to the vehicle so as to operate as a brake light.

5. The light source of claim 1, the optical attenuator structure being arranged to reflect at least a portion of the output light propagating in the second direction to propagate in the first direction.

6. The light source of claim 1, the optical attenuator structure substantially continuously spanning the entire areal extent of the set of light-emitting elements.

7. The light source of claim 6, the optical attenuator structure being arranged as a partially reflecting and partially transmitting reflector across the visible spectrum.

8. The light source of claim 7, the optical attenuator structure being at least 10% reflective and at most 90% reflective.

9. The light source of claim 1, the microLEDs comprising UV- or visible-emitting, direct-emitting or phosphor-converted, semiconductor microLEDs, the microLEDs including one or more materials among III-V, II-VI, or Group IV semiconductor materials.

10. The light source of claim 1: (i) each light-emitting element having a largest transverse dimension that is less than 200 μm , (ii) the light-emitting elements occupying less than 25% of the areal extent of the set, or the set of light-emitting elements occupying an area of the substrate having a smallest transverse dimension that is greater than 5 mm.

11. The light source of claim 1, each light-emitting element of the set including only a single microLED.

12. The light source of claim 1, each light-emitting element of the set being arranged and connected so as to be operable independently of at least one other light-emitting element of the set.

13. The light source of claim 1, each light-emitting element of the set including multiple microLEDs, each microLED of a given light-emitting element being arranged and connected so as to be operable independently of at least one other microLED of that light-emitting element.

14. The light source of claim 1 further comprising multiple electrically conductive traces on or within the substrate that are arranged and connected for providing electrical drive current to the light-emitting elements of the set, the traces being sufficiently transparent, sufficiently narrow, or spaced sufficiently far apart so as to enable visual observa-

tion of the scene through or reflected by the substrate along the sight line that passes through the set of light-emitting elements and the set of electrically conductive traces.

15. The light source of claim 1, each light-emitting element being sufficiently small in at least one transverse dimension, and the light-emitting elements of the set being spaced sufficiently far apart, so that the set does not substantially interfere with visual observation of the scene through the substrate by a naked eye of a human observer along the sight line that passes through the set.

16. The light source of claim 1, each light-emitting element being sufficiently small in at least one transverse dimension, and the light-emitting elements of the set being spaced sufficiently far apart, so that to a naked eye of a human observer the set is only negligibly visible with the light-emitting elements in an off-state.

17. A method comprising, with a substrate positioned in a sight line along which a scene is observed through the substrate by an observer, operating a set of multiple light-emitting elements positioned on or within the substrate to emit visible output light,

the substrate being transparent for visible light,

each light-emitting element comprising one or more inorganic microLEDs that are arranged to generate and emit the visible output light to propagate primarily toward the scene out-of-plane relative to a corresponding localized area of the substrate around that light-emitting element,

an optical attenuator structure positioned on or facing a surface of the substrate that faces the observer,

each light-emitting element of the set being sufficiently small in at least one transverse dimension, and the light-emitting elements occupying a sufficiently small fraction of an areal extent of the set, so as to enable visual observation of the scene through the substrate along the sight line, the sight line passing through the set of light-emitting elements, and

the optical attenuator structure being positioned and arranged so as to (i) attenuate or block transmission therethrough of a portion of the visible output light that propagates toward the observer, and (ii) transmit sufficient visible light so as to enable visual observation of the scene along the sight line through the optical attenuator structure and through the set of light-emitting elements.

18. The method of claim 17, the one or more microLEDs emitting output light a vacuum wavelength λ_0 , the optical attenuator structure being arranged as a short-pass, long-pass, or notch optical filter that attenuates or blocks transmission of output light at the vacuum wavelength λ_0 .

19. The method of claim 17: (i) the optical attenuator structure substantially continuously spanning the entire areal extent of the set of light-emitting elements, (ii) the optical attenuator structure being arranged as a partially reflecting and partially transmitting reflector across the visible spectrum, and (iii) the optical attenuator structure being at least 10% reflective and at most 90% reflective.

20. The method of claim 17: (i) the substrate being a rear window of a motor vehicle with the observer being inside the vehicle and the scene being outside the vehicle, (ii) the visible output light being red output light, (iii) the optical attenuator structure being arranged as a short-pass or notch optical filter that attenuates or blocks transmission of the red

output light into the vehicle, and (iv) the light source being arranged and coupled to the vehicle so as to operate as a brake light.

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